

# **OphDDI-CDS: An Absorption-First Deterministic Cascade for Ophthalmic Drug-Drug Interaction Screening**

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## Abstract

**Background.** Ophthalmic topical medications reach clinically meaningful systemic exposure, yet general-purpose drug-drug interaction (DDI) databases grade severity from systemic administration and do not model absorption from the topical route. Whether that assumption holds at registry scale, and whether evidence-graded assessment agrees with prescribing clinicians, is unknown.

**Results.** We built OphDDI-CDS, a deterministic three-layer cascade (40 mechanism rules, each with a source record; an absorption-scaled class matrix; a hard-constraint layer) over 113 ophthalmic and 232 systemic agents, and compared three ways of assigning ophthalmic–systemic DDI risk. Against DDInter severity for all 1,933 registry-matched pairs, 75.3% were graded lower once absorption was modelled (352 downgraded from high), and 80.5% of downgrades were attributable to low systemic absorption. On a 118-case literature-derived gold the cascade reached 0.907 exact accuracy (kappa 0.857) versus 0.415–0.737 for three LLM baselines. Re-run with a reasoning model and a 3000-token budget, the baselines’ label agreement rose (0.576–0.737) but high-risk sensitivity collapsed to 0.214–0.536, a conservative rarely-high-risk policy; the cascade retained 0.964. On a semi-independent 92-case audit set the cascade reached 0.674 accuracy with high-risk sensitivity 0.320 (8 of 25); restricted to the 62 cases whose severity is free of the system’s own absorption assumption, accuracy is 0.548 (34 of 62). One adequate-budget baseline scored higher on both (0.728 and 0.400),

the only such comparison. Two independent ophthalmologists reviewed 12 assigned grades with both the proposed gold and the system grade visible; they agreed closely with each other (kappa 0.840) yet did not endorse the assigned grades, rating 10 of 12 cases one tier lower; because the questionnaire displayed both grades and the system reproduced its gold, this exercise is not an independent validation and is reported as such.

**Conclusions.** The same ophthalmic–systemic pair is assigned different risk by systemic-route databases, evidence-graded rules and clinical intuition, and the first of these disagrees with absorption-aware assessment for three of every four registry-matched pairs. An absorption-first deterministic cascade makes those disagreements enumerable, individually traceable, and auditable; its false-negative profile and clinical value require prospective, outcome-linked calibration, and the literature-derived gold bounds the strength of all accuracy claims reported here.

## 1. Background

Ophthalmic topical medications are among the most widely prescribed drugs worldwide, yet their systemic absorption is frequently underestimated in routine DDI screening. Although eye drops are assumed to act locally, several agents achieve clinically meaningful systemic exposure. Non-selective beta-blockers such as timolol reach plasma concentrations comparable to low oral doses through nasolacrimal drainage and mucosal absorption; Alvan et al. measured plasma concentrations after ocular administration [1], and Rait notes that systemic absorption occurs in relatively small amounts but is clinically consequential in susceptible patients [2]. (An earlier version cited these two references for an “approximately 80%” bioavailability figure; neither reports it. That

order of magnitude is supported by Korte et al. [N1], cited below.) Topical carteolol can likewise cause atrioventricular block [3], and severe systemic reactions to ophthalmic timolol have been linked to CYP2D6 poor metabolizers [4]. Severe systemic events are documented: topical timolol combined with verapamil can cause bradycardia and cardiogenic shock [5], and further case series of timolol-associated syncope have been reported [6,7]. Ageing patients are particularly vulnerable, including respiratory and cardiac complications of ocular beta-blockade [8]. Alpha-2 agonists can produce CNS depression, particularly in children [9-12], and pediatric patients are generally more susceptible to systemic effects of ocular medications [13]. Nasolacrimal occlusion or punctal compression measurably reduces the systemic exposure from instilled drops [14]. In elderly patients, who are often prescribed multiple systemic medications, such additive effects can accumulate into clinically significant harm [6-8,15].

General-purpose DDI databases, including DDInter, provide comprehensive pair-level severity ratings derived predominantly from systemic administration [16]. These resources are of limited direct utility for ophthalmic prescribing. DDI screening methods have long been recognized as clinically relevant but methodologically heterogeneous [17], and automated detection in practice remains incomplete [18,19]. Most importantly, general-purpose databases do not encode the decisive ophthalmic pharmacokinetic modifier, namely the magnitude of systemic absorption from topical administration, and they omit dose-route corrections derived from ophthalmic pharmacology. Route awareness is not wholly absent from mainstream resources: Drugs.com maintains a separate ophthalmic entity for timolol with its own interaction list and severity distribution, and Lexicomp asks the user to specify the route of administration. What is

absent is not route awareness as such but an explicit, registry-wide absorption model that would let a prescriber see how a systemic-route severity rating changes once topical absorption is taken into account; that is the gap this work addresses. As a result, the same drug pair can be rated high-risk for oral administration while being clinically negligible for topical use. The magnitude of this mismatch has not been measured. It is acknowledged qualitatively that route matters, but no study has quantified, at registry scale, how often a systemic-route severity rating changes once absorption from the topical route is modelled; nor has an evidence-graded ophthalmic assessment been compared with the judgment of prescribing clinicians for the same pairs. Clinicians therefore lack both a mechanism-level ophthalmic-specific screening tool and any estimate of how far the interaction alerts they already receive depart from ophthalmic pharmacology.

Recent LLM approaches to clinical DDI screening offer flexibility but raise reliability concerns. Unconstrained models can hallucinate interactions and are non-reproducible, and retrieval-augmented variants inherit the biases of their retrieval corpora. In our own evaluation, retrieval-augmented variants captured only part of the gap to a purpose-built cascade, and the ranking of retrieval methods depended on how much output budget the model was given [20,21]; Section 3.6 reports both settings. Systematic reviews of computerized clinical decision support show modest but variable effects on practice [22,23], and high alert-override rates remain a central adoption problem [24,25].

Moreover, LLM outputs are non-deterministic and difficult to audit in a clinical context.

We present an ophthalmic DDI decision-support system built on a deterministic, fully explainable three-layer cascade. The L1 rule engine contains 40 mechanism rules, each

carrying at least one source record. The L2 drug-class matrix applies absorption-scaled upgrades. The L3 hard constraint encodes ophthalmic systemic-absorption pharmacology. The system is grounded in a curated knowledge base of 1,307 verified evidence chunks (113 ophthalmic agents, 232 systemic agents). It produces traceable risk grades (high, medium, or low) with matched rule IDs and evidence chains. The production path requires zero LLM calls, shows zero variance across temperature conditions, and completes inference in milliseconds.

Our contributions are fourfold. First, we quantify the divergence rather than assert it: across all 1,933 ophthalmic–systemic pairs for which DDInter provides a severity rating, absorption-aware assessment grades 75.3% of pairs lower, 352 pairs move down from high, and a single nameable mechanism — low systemic absorption from the topical route — accounts for 80.5% of all downgrades. This turns a known qualitative caveat into a measured property of the database. Second, we provide the instrument that makes the measurement possible: a reproducible, source-traceable rule cascade for ophthalmic DDIs, including the class-matrix cells and absorption scaling absent from general-purpose databases, whose every decision returns the rule identifiers and evidence chain behind it. Third, we show that the divergence is not one disagreement but three, by comparing the same pairs under systemic-route severity, evidence-graded assessment, and expert grade review; on the 12-case expert review the experts disagreed with the assigned grades about one tier lower on 10 of 12 cases, and the same near-zero agreement appears against both the system and its gold because the system reproduced that gold. Fourth, we report the evaluation in a form that can be audited and re-run: a four-phase protocol (rule consistency, development-phase validation, a semi-independent external audit, and a

clinical grade review (not an independent validation; see Section 4.4)) with all four datasets reported, including the two on which the system performs poorly, and a downloadable zero-configuration web interface for clinical and educational use. The system is positioned as a mechanism-based second reader, not as a replacement for clinical judgment.

## 2. Methods

### 2.1 Study design and data sources

We developed and evaluated a clinical decision-support system for DDIs between ophthalmic topical medications and systemic drugs. The system combines a deterministic, fully explainable rule cascade with a curated ophthalmic DDI knowledge base.

Evaluation followed a four-phase protocol: rule consistency, development-phase validation, a semi-independent external audit against DDInter, and an independent two-expert clinical assessment, comparing the system with three LLM baselines under five temperature conditions.

**How route of administration is represented.** Route is modelled explicitly, through two mechanisms rather than through a route field. First, the ophthalmic registry is restricted to agents administered to the eye and each carries a systemic-absorption tier (high, medium, low, very-low, minimal or none); that tier is the operational proxy for how much of a topical dose reaches the systemic circulation, and it is what the absorption-scaling step acts on. Second, the rule layer encodes route-dependent predicates directly, for example a rule that requires the flag `systemic_absorption_high` before it fires. Route is therefore not a free-text attribute of a pair but a numeric modifier and a flag that gate which rules and

matrix cells can assign a grade. The tier assignment procedure and its sources are described in Section 2.2 and listed per agent in Supplementary Table S5.

Evidence and gold labels were derived from multiple sources. DDInter 2.0, a curated database of 302,516 DDI records [N2]. FDA prescribing labels for ophthalmic products provided product-specific warnings. Stockley’s Drug Interactions and the British National Formulary [26] provided mechanism-level annotations, and peer-reviewed literature was retrieved from PubMed. All 1,307 evidence chunks were filtered to approved-release status and graded by evidence level (A, regulatory label; B, meta-analysis or systematic review; C, original study; D, in vitro, animal, or pharmacokinetic first principles; E, class-level mechanism inference). The approved distribution was A = 53, B = 191, C = 215, D = 584, E = 264.

## 2.2 System architecture and knowledge base

The production system (full\_system) is a three-layer deterministic cascade with zero LLM calls (Figure 1).

**L1 rule engine.** Forty mechanism rules define DDI risk, including additive beta-blockade, beta-blocker–calcium-channel-blocker additive effects, additive alpha-2 CNS depression, CYP3A4 induction or inhibition, and QT-prolongation risk with low-absorption fluoroquinolones. Each rule is defined by entity-flag predicates with match-all or match-any semantics, and the highest-severity hit is retained (MAX-MATCH). The 43 policy entries comprise 40 rules plus 3 patient-factor policies; Table 2 lists the rule inventory with flags and match semantics. **No second-order interactions are modelled:** each rule is evaluated independently on the properties of the two agents, and no rule’s activation

depends on whether another rule fired. The only cross-layer composition is the final maximum of the rule and matrix layers (Section 4.2); combination products are handled by assigning the tier of the dominant component rather than by modelling component interactions.

Citation coverage is narrower than the phrase “citation-verified” would suggest, and we state it precisely. Every rule carries at least one source record, but the records are heterogeneous: 41 are DrugBank entries, 7 are regulatory label records, 2 are DOIs, 2 are DDInter pair annotations and 1 is a PubMed citation. Independent verification against three sources (Crossref, OpenAlex, PubMed) with retraction screening was performed for a subset: the verification report covers 8 manifest items spanning 7 distinct DOIs, and of these, three rules resolve to a verified literature citation (additive\_beta\_blockade, beta\_blocker\_ccb\_additive, beta\_blocker\_digoxin\_additive). The remaining rules cite DrugBank or label records, which were not independently re-verified in this work. The report is bound to a SHA-256 of the rule configuration so that any later edit invalidates it, and it is deposited with the manuscript (Supplementary Text S2).

Two consequences of that partial coverage should be noted. First, several rules whose output class would call for a citation declare a citation list that does not resolve to a verified item; the gate accepts this, and the claim in earlier versions of this manuscript that all forty rules were citation-verified was overstated. Second, beta\_blocker\_digoxin\_additive cites a study of additive topical-plus-systemic beta-blockade rather than a digoxin-specific source, and it grades as high a pair that its own DDInter annotation records as moderate; we retain the conservative high grade but flag the provenance gap here.

**L2 class matrix.** A drug-class matrix relates ophthalmic drug classes to systemic drug classes. The released configuration declares 21 ophthalmic row keys and 40 systemic column keys, of which 34 columns are used, and assigns a level to 104 cells, of which 83 carry an actionable grade (high/medium/low) and 21 are explicitly marked “none” and ignored by the engine; cells that are not defined fall back to a default cell. (Earlier versions of this manuscript described the matrix as  $20 \times 37$  with 82 cells; those figures did not match the configuration and are corrected here.) Four upgrade rules require both a class match and specific absorption or selectivity flags, for example a non-selective beta-blocker with high systemic absorption. An absorption-scaling step forces agents with low or very-low systemic absorption to low risk unless an upgrade condition applies; high or medium absorption preserves the cell level. This step is **not empirically validated** against outcome data: it is an author-specified design decision grounded in the pharmacokinetic principle that a dose which does not reach the systemic circulation cannot produce a systemic interaction, and its behavioural consequence is quantified in Section 3.4 (which reports how many registry pairs change grade because of it) and in the deposit’s path-attribution audit. Where the tier assignment is uncertain, the pair is conservatively left to the rule layer rather than forced low. Note that this forcing applies within the matrix layer only; the final grade is the maximum of the matrix layer and the rule layer, as discussed in Section 4.2. Most matrix cells are therefore deliberately undefined: the matrix is sparse by design, encoding the 83 graded class pairs for which a mechanism or an absorption-based expectation exists (21 further cells are marked “none” to record that the pair was reviewed and needs no action), and any pair that reaches neither a rule nor a defined cell falls through to the L3 default-low path rather than

receiving an interpolated grade. **Layer scope of the class mapping:** the drug-to-class mapping (113 ophthalmic agents, 232 systemic agents; configs/drug\_classes.yaml) is consumed by the L2 matrix layer only. The L1 rules operate on mechanism flags rather than classes, and the L3 hard constraint operates on the absorption tier alone.

**Which layer produces the grade.** Because the cascade returns the maximum of the rule and matrix layers, the layer that determines each grade is worth stating. Across all 26,216 registry pairs the class matrix determines 68.7%, the G19 hard constraint 12.2%, the rule layer 12.0%, and an explicit default-low path 7.1% for pairs with no rule or matrix signal; these four paths are disjoint and their outputs are recorded in the audit deposited with this manuscript. The rule layer is not absorption-gated, which has one consequence we report rather than leave implicit: 202 pairs whose ophthalmic agent carries a low-or-below absorption tier receive medium (107) or high (95) grades, and every one of them arrives through the rule path. All 95 high grades come from a single rule, nsaid\_antihypertensive\_antagonism, which requires no absorption flag. Section 4.2 quantifies that consequence and why we did not gate the rule.

**L3 hard constraint.** Low or very-low absorption with no rule hit and no matrix signal yields low risk (G19). For pairs with no rule or matrix signal, the default is low with a documented decision path. Unknown or ambiguous medications are not silently classified: the engine returns an explicit unknown-drug path and the screening service raises an error requesting resolution, so out-of-coverage pairs are never assigned a risk grade by default.

**Drug-class mapping** (113 ophthalmic agents, 232 systemic agents) was built from ATC codes, DrugBank records, and manual pharmacologic review, with DrugBank identifiers for 103 and 218 agents respectively; combination products were tiered by their dominant

component. **Absorption tiers and their definition.** Ophthalmic systemic-absorption tiers were assigned from pharmacokinetic sources and regulatory labels, including nasolacrimal-drainage bioavailability for timolol [1], plasma-level measurements for timolol [27], and instillation-technique effects on systemic exposure [14]. The registry uses **six** tier values, not the four used in earlier descriptions of this work: systemic\_absorption\_high, systemic\_absorption\_medium, systemic\_absorption\_low, systemic\_absorption\_very\_low, minimal\_systemic\_absorption and systemic\_absorption\_none (113 agents: 15 high, 17 medium, 61 low, 9 very-low, 7 minimal, 4 none). The L3 hard constraint fires on two of these values, systemic\_absorption\_low and systemic\_absorption\_very\_low, as implemented in kg\_layer.py; the remaining four values preserve the assigned level. (Section 2.2 and Section 4.2 of the previous revision stated that four tiers were tested; the code tests two. The final grades are unaffected because the other low-exposure values are already mapped to low risk within the matrix layer.) The tier is a **categorical assignment, not a measured quantity**: the intended reading is fractional systemic bioavailability after topical instillation (high above roughly 50%, medium roughly 10-50%, low below roughly 10%, with very-low, minimal and none indicating progressively smaller exposure), but the sources report heterogeneous quantities (bioavailability fractions, plasma concentrations, or qualitative statements), so the tiers should be read as an explicitly stated ordinal model rather than as measured exposure for every agent. Two limitations follow and are stated here rather than discovered later: first, the current assignment contains at least one internal inconsistency that has now been corrected (flurbiprofen was tiered high in the registry while the curation file recorded low for the same evidence sources; it is tiered low as of this

revision), and second, nine of the 113 agents in the ophthalmic registry are not topically administered (six intravitreal anti-VEGF agents, intraocular acetylcholine, and two surgical viscoelastics), so the topical-absorption rationale does not apply to them in the same way. Both are flagged as required corrections in Section 4.4; the per-agent table is Supplementary Table S5. **The table’s source column does not evidence the tiers individually:** all 113 rows record the same string (DrugBank DBxxxxx|PubMed, with PubMed as a bare word and no identifier), and individual source records are named in this section for timolol [1,27] and for atropine and dexamethasone [29] only. The remaining tiers are the study team’s curation assignment; Table S5 marks this explicitly so the support behind each row is not overstated. Ocular drug transfer from systemic administration is likewise complex and incompletely characterized [28]. For atropine and dexamethasone, tier assignment drew on regulatory labels and pharmacovigilance evidence [29]. The full per-agent tier table, with a source record for each agent, is provided as Supplementary Table S5.

The evidence base (1,307 chunks) was embedded with a 2048-dimensional Doubao embedding model (doubao-embedding-vision-251215) and indexed with LightRAG [21] (working directory data/processed/lightrag\_index; 1,307 documents, 1,627 entities, 1,836 relations). Index freshness was verified against evidence fingerprints. Retrieval baselines used BM25 (Best Matching 25) [20] over the same evidence chunks.

### 2.3 Validation datasets and evaluation protocol

L1 literature-derived gold (118 cases). Gold labels were constructed by the study team from authoritative sources, without reference to system outputs, following the data-selection principle set out in Section 2.1. **Selection criteria.** The set was assembled to

span the dose-route correction the system models rather than to sample representatively: it combines every FDA-label mechanism annotation that met the inclusion rule, a procedurally frozen blind subset, and guideline-sourced cases chosen to cover classes where the systemic route and the ophthalmic route diverge most. Specifically, the 20 mechanism annotations are those FDA-label, Stockley's and BNF statements that specify a mechanism rather than only a severity; the 58 guideline-sourced cases were selected from FDA product labels, BNF interaction tables and Stockley's to include at least one pair per ophthalmic class in the registry, with case-level source references recorded in the gold file. **To be explicit about who constructed the gold:** the gold was constructed by the two authors of the original submission (GW and YX), who performed the curation and annotation. The two authors added during revision (HL and WS) contributed clinical input and interpretation of the results and reviewed the manuscript, but did not construct the gold and did not take part in the expert grade review. "No human expert participated in gold construction" therefore means that no *independent* expert was involved; none of the expert opinion used here was blinded to the system. **How the gold shares sources with the rules:** the sharing differs by subset — the 20 mechanism annotations and the 58 guideline-sourced cases were labelled from the same FDA/Stockley's/BNF statements that the rules cite, whereas the 40 frozen blind cases were labelled by the study team from the same literature without case-by-case reuse of a rule citation. **Contradictory evidence:** where sources disagreed we retained the more conservative (higher) grade and recorded the disagreement in the case file; the ten cases where our label still differs from the guideline are listed with their rationale in Supplementary Table S1. We further disclose that one of the two 20-case blind subsets (blind 2, cases C01–C20) entered the

development loop through error attribution during class-matrix versions v2.1–v2.3; the other subset (blind 1, cases B01–B20) was never used in development. Gold-level distribution: 28 high, 51 medium, 39 low (Table 4; all four datasets as Supplementary Table S8).

DDInter audit set (92 cases). A stratified random sample by absorption tier and DDInter severity. **Correction rule.** A case was corrected from the DDInter severity to low when the ophthalmic agent’s registry absorption tier is low, very low, minimal or none, no mechanism rule fires for the pair, and no class-matrix upgrade applies; under those three conditions the systemic-route severity cannot be reached at ophthalmic doses. Thirty of 92 cases met the rule; the remaining 62 were kept at the DDInter level. The audit is therefore semi-independent: the severity component is third-party, but the absorption correction shares the system’s central assumption for 30 cases.

Development-phase and gold-calibration reference datasets (reported separately from the validation sets above). These three sets were used as development gates, not as validation: a development set (200 cases, DDInter-derived), an external PubMed-co-mention set (70 cases), and an internal validation set (78 cases). Their features are identical to the validation sets, namely the ophthalmic agent, the systemic agent, the agent-level flags and classes used by the rule and matrix layers, and a three-level risk label; the labels differ in provenance. The development and external sets carry unadjusted systemic-route gold, and the external set’s co-mention gold was audited as noisy (literature co-mention is frequently unrelated to the pair). Results on both are reported in Section 3.2 with the explicit caveat that they quantify the ophthalmic adjustment rather than performance. The

internal set served as the development gate; 78 of 118 L1 cases overlap it by case identity (gold independent, cases not entirely novel).

## 2.4 Baselines and evaluation protocol

Four methods were compared: `pure_llm` (a single LLM call with a domain-general system prompt and no retrieved evidence), `naive_rag` (BM25 retrieval plus LLM), `lightrag` (LLM with knowledge-graph context), and `full_system` (deterministic cascade). The three LLM methods shared an identical system prompt containing the three-level risk definitions and a general ophthalmic-absorption reminder; the exact prompts are provided as Supplementary Text S3. **Retrieval protocols, in full.** `naive_rag` performs BM25 retrieval with `rank_bm25.BM25Okapi` ( $k1 = 1.5$ ,  $b = 0.75$ ) over the same 1,307 evidence chunks that populate the index, lowercased and whitespace-tokenised on both sides; the query is the string “ ophthalmic interaction”, i.e. the drug names are included, the top 3 passages are retained, each truncated to 400 characters, and they are prepended to the case as an Evidence: block. `lightrag` instead prepends a Drug profiles: block containing the knowledge-graph profiles of both agents, rendered by the curated `PreciseKG.full_kg_context()` function. That function reads the registries and rule configuration directly rather than querying the LightRAG vector index, so this is a knowledge-graph context baseline and not a vector-retrieval baseline; we describe the mechanism accurately and retain the name for continuity with earlier protocol versions. The exact prompts for all three LLM methods, the BM25 parameters, the parse rule and the temperature schedule are given in full in **Supplementary Text S3** and in the deposit file `configs/prompts.yaml`. LLM baselines used the DeepSeek API model identifier `deepseek-chat`, accessed through the OpenAI-compatible endpoint, with an output cap of

10 tokens and no chain-of-thought prompting. The endpoint does not expose the underlying parameter count or checkpoint revision, so we report the identifier as used and note that this is a single proprietary endpoint rather than a locally pinned architecture.

**Input language was English throughout**, for both the system prompt and every case, in the form Drug pair: ophthalmic X (eye drops) + systemic Y (oral/IV). Temperature was 0.0 for condition 0 and 0.3, 0.6, 0.9, 1.2 for conditions 1–4. The 10-token cap is a deliberate design choice that forces a one-word answer; it removes any possibility of reasoning or justification, and this bounds what the comparison can support (Section 4.2). We refer to the five conditions as temperature conditions rather than seeds, because only temperature varied; the deterministic method is identical across conditions by construction. Outputs were parsed by substring matching with an explicit unknown fallback; no non-conforming outputs occurred on the L1 or audit sets. BM25 used the rank\_bm25 implementation ( $k1 = 1.5$ ,  $b = 0.75$ ,  $\text{top-k} = 3$ ). Because the 1,307-chunk corpus also informed the gold labels, the retrieval baselines access gold-source material, which the deterministic system does not use; this asymmetry is disclosed and gives the baselines, if anything, an advantage. Five conditions  $\times$  four methods  $\times$  four datasets produced 9,600 prediction rows with zero API failures. Supplementary Text S5 records the run configuration and the file-level fingerprints needed to verify that the deposited predictions reproduce these figures. **Reasoning-model re-run (reported in Section 3.6 and Table 3)**. The three LLM baselines were additionally re-run under an adequate output budget with a reasoning model. Setup, stated in full: provider Zhipu (Zhipu AI, China) through the coding-plan endpoint <https://open.bigmodel.cn/api/coding/paas/v4>, model identifier glm-5.3-flash, temperature 0.0, max\_tokens 3000, five repeated samples per case,

temperature held fixed so the repetition captures sampling variance rather than a temperature schedule. **Reasoning tokens count against the 3000-token budget:** the model emits reasoning\_content before content, so the budget covers both. **Zero responses were truncated** at this budget (completion\_tokens never reached max\_tokens across all 3,150 calls). Split votes were resolved by majority over the five samples; the deposited parse rule is applied to each sample first (exact match, then substring high > medium/moderate > low, else unknown), and no sample returned unknown. The run was executed on 23–24 September 2026 and produced 3,150 raw responses, deposited in full at data/predictions/glm53\_baselines/ with the run script scripts/rerun\_glm53\_baselines.py. **Single-model caveat:** this comparison uses one commercial reasoning model at one endpoint; the endpoint does not expose checkpoint revision, so the finding that reasoning models adopt a conservative rarely-high-risk policy is a property of this model family at this budget, not a claim about reasoning models in general. The deterministic cascade requires no model call and is unchanged between the two settings.

Metrics. Exact three-level accuracy (Acc3; high/medium/low), high-risk sensitivity and specificity (high versus non-high), high-risk F1, and Cohen’s kappa [30] with Landis–Koch interpretation [31]. Method differences were tested with the exact two-sided McNemar test on discordant case pairs (full\_system versus each baseline; condition 0). Confidence intervals: Wilson intervals for proportions, Clopper–Pearson intervals for perfect subsets, and nonparametric bootstrap intervals (2,000 resamples) for kappa. Standard deviations across temperature conditions are population SDs. No power calculation was performed; the study is a system-development report rather than a confirmatory trial, and all intervals are reported as descriptive uncertainty.

## **2.5 Expert grade review**

Two independent ophthalmologists reviewed the assigned grades for 12 representative cases under a grade-review protocol (Supplementary Table S3; the instrument, with the experts' recorded judgments, is provided in the deposit). For each case the instrument displayed the drug pair, a short clinical scenario, the proposed gold grade, and the system's grade together with a one-line mechanistic rationale; the expert then recorded their own judgment (high, medium, or low) on the same three-level scale. Because both the proposed gold and the system grade were visible, this exercise is a grade review and not a blinded independent assessment, and the agreement statistics below are reported accordingly. Cases were selected by the study team to span errata confirmations, upgrade disputes, downgrade confirmations, potency-granularity boundaries, and typical cases. Mechanism-level evidence and references were not provided, and the experts were asked to judge from clinical practice rather than from per-case mechanism review. Ratings were collected on paper (12 cases across two pages), scanned, and transcribed by the study team; expert 1's transcription was verified against the scanned questionnaire, and expert 2's ratings were transcribed by the corresponding author from the expert's written responses. Inter-rater reliability and expert–system and expert–gold agreement were summarized with Cohen's kappa (unweighted and quadratic-weighted), with the maximum achievable unweighted kappa given the marginal distributions reported as context.

## **2.6 Intended use and regulatory status**

The system is intended as a research-grade, mechanism-based second reader for clinicians and pharmacists screening ophthalmic–systemic combinations. It is not

intended to replace clinical or pharmacy judgment, and its outputs require clinical review. Out-of-coverage medications are rejected with an explicit error rather than scored. The shipped web interface carries a disclaimer to this effect. The system is not marketed as a medical device; regulatory classification for prospective clinical deployment would require formal assessment under applicable frameworks.

## **2.7 Software and reproducibility**

The system is implemented in Python 3.11 (package `ophthalmic_ddi_cds_agent`). The complete source code, the curated knowledge base, the rule-citation verification tooling and the evaluation scripts are publicly available at <https://github.com/wangguoqing932/ophddi-cds>, together with a synthetic demonstration dataset and a one-command demonstration entry point. Engine, rules and gold versions are pinned by SHA-256 fingerprints. Because a file's digest depends on its line endings, the fingerprints are defined over **LF-normalised content**: a reader should convert CRLF to LF before hashing, otherwise a Windows checkout and a Linux checkout of the same file yield different digests (the deposited configs/rules.yaml, for example, hashes to 2afabaad81161c40 under the original CRLF line endings but to b5c77f6fde8a9f7d after LF normalisation). The evaluated version was rules 2afabaad81161c40 (CRLF as deposited); the evidence and index fingerprints are recorded alongside the deposited artefacts rather than only in the project log. All statistics reported here are reproducible from the released prediction files via the provided scripts. The report follows the TRIPOD-AI reporting guidance for prediction-model studies where applicable; a completed self-assessment checklist is provided as Supplementary Text S6. The system is a rule-based decision-support tool rather than a statistical prediction model, and several

TRIPOD-AI items (for example model-updating procedures) are not applicable; this is stated per item in the checklist. Use of generative AI tools is documented here as the journal requires: DeepSeek-chat was used for the LLM-baseline evaluations and as an assistive tool in knowledge-engineering tasks (registry curation, evidence grading, and gold-label annotation), and Zhipu GLM-4.6V was used for visual verification of scanned expert questionnaires and figures. No AI tool is listed as an author, and all AI-assisted outputs were reviewed and validated by the named authors (see Declarations).

### **3. Results**

#### **3.1 System construction**

The knowledge base comprises 113 ophthalmic topical agents and 232 systemic medications (Table 1; machine-readable form as Supplementary Table S4), 1,307 approved evidence chunks ( $A = 53$ ,  $B = 191$ ,  $C = 215$ ,  $D = 584$ ,  $E = 264$ ; composition detailed in Supplementary Text S1), and a LightRAG index over 1,307 documents. The rule engine comprises 40 mechanism rules plus 3 patient-factor policies (43 policy entries), each carrying at least one source record, with zero policy errors (Table 2; full inventory as Supplementary Table S6); the coverage of those records, and its limits, are stated in Section 2.2. The class-matrix layer defines 104 cells across 21 ophthalmic row keys and 40 systemic column keys (34 columns in use) with absorption-scaled upgrades. Figure 4 illustrates representative system outputs for a curated set of clinical demo cases.

#### **3.2 Development-phase and gold-calibration reference results**

The deterministic engine achieved 100% rule consistency on the development dataset (Phase 1). On the development-phase internal validation set (78 cases, gold synced to the

corrected blind labels), sensitivity was 1.000 and specificity 0.984; we report these as a development gate, not as independent evidence, because 78 of 118 L1 cases overlap this set by case identity.

Against the unadjusted systemic-route gold of the development set (200 DDInter-derived cases), the system achieved 0.210 exact accuracy (sensitivity 0.069, specificity 0.895); against the external PubMed co-mention gold (70 cases, audited as noisy), it achieved 0.443 (sensitivity 0.077, specificity 0.935). Under the evaluation protocol of Section 2.3, these low agreement values are expected and informative: they quantify the magnitude of the system’s ophthalmic absorption correction relative to systemic-route labels, and they are not interpretable as performance. We report them to complete the record of all four evaluated datasets.

### **3.3 Evidence-graded agreement and comparison with LLM baselines (118 cases)**

On this set the gold and the rules draw on the same literature (Section 3.9), so agreement measures the internal consistency of an explicitly evidence-graded encoding rather than accuracy against an independent standard. What the set does support is a like-for-like comparison of four methods on identical cases, and the ability to inspect every disagreement individually. Table 3 and Figure 2 summarize the comparison (five temperature conditions each; mean  $\pm$  SD), and Supplementary Table S7 gives the same figures dataset by dataset and budget by budget. The state-of-the-art LLM baselines acting as comparators receive the same three-level risk definitions and the same general absorption reminder in their system prompt (Section 2.4).

The deterministic system (full\_system) achieved 0.907 exact accuracy (107/118; Wilson 95% CI 0.841–0.947), 0.964 high-risk sensitivity (27 of 28; CI 0.823–0.994), 0.989 specificity, and Cohen’s kappa 0.857 (bootstrap CI 0.772–0.934), with zero variance across conditions. Lightrag reached 0.737 accuracy, 0.779 sensitivity, and 0.936 specificity (kappa 0.603). Pure\_llm reached 0.475 accuracy and 0.771 sensitivity (kappa 0.256). Naive\_rag reached 0.415 accuracy and 0.757 sensitivity (kappa 0.182). Exact two-sided McNemar tests (condition 0) were significant for full\_system versus each baseline: versus pure\_llm (54 discordant pairs in favor of the system, 2 against;  $p = 4.4 \times 10^{-14}$ ), versus naive\_rag (60 versus 4;  $p = 7.4 \times 10^{-14}$ ), and versus lightrag (20 versus 1;  $p = 2.1 \times 10^{-5}$ ).

The accuracy gradient full\_system > lightrag > pure\_llm > naive\_rag was stable across all five temperature conditions on this set. The accuracy gap between full\_system and the best LLM baseline (lightrag) was 0.170, indicating that the deterministic knowledge engineering contributed more than retrieval augmentation alone. In this 10-token setting naive\_rag scored below pure\_llm, but that ordering does not survive an adequate output budget (Section 3.6), so we do not interpret it as a property of keyword retrieval.

In the per-subset analysis, the procedurally frozen blind subset ( $n = 40$ , SHA-locked) reached 0.975 accuracy (39 of 40; Clopper–Pearson CI 0.868–0.999); the single miss is flurbiprofen  $\times$  warfarin, discussed in Section 3.5. The FDA/Stockley’s/BNF mechanism subset ( $n = 20$ ) reached 1.000. The guideline-sourced subset ( $n = 58$ ) reached 0.828 (48/58). Eleven exact-scale disagreements remain in total: ten in the guideline subset and the flurbiprofen pair in the frozen subset. The guideline disagreements concern judgment boundaries, including QT risk with low-absorption fluoroquinolones, CYP3A4 inhibition

with low-absorption macrolides or azoles, weak-corticosteroid potency granularity, and anticholinergic overlap with tricyclic antidepressants. Each disagreement carries a documented pharmacological rationale (Supplementary Table S1).

### **3.4 Registry-scale divergence between systemic-route severity and absorption-aware assessment**

The largest result in this study does not depend on the gold labels, because both sides of the comparison are pre-specified: DDInter supplies a systemic-route severity rating, and the cascade supplies an absorption-aware grade. We therefore compared every ophthalmic–systemic pair in the  $113 \times 232$  registries for which DDInter 2.0 provides a severity record — 1,933 pairs. Of these, 1,455 pairs (75.3%) were graded lower by the system, including 352 pairs downgraded from high (37 to medium, 315 to low; 18.2% of all pairs) and 1,103 from medium to low; 136 pairs (7.0%) were upgraded. Of all downgrades, 80.5% (1,171 of 1,455) were attributable to low systemic absorption. In other words, three of every four pairs that carry a severity rating would be presented to a prescriber differently under an absorption-first assessment, and the majority of that shift follows from one pharmacokinetic mechanism rather than from many independent judgements.

Two caveats bound this measurement. First, it quantifies divergence, not error: the analysis shows that the two perspectives rarely coincide for the topical route, not that the database ratings are wrong. Second, the absorption tiers that drive the downgrades were assigned by the study team from pharmacokinetic sources and regulatory labels (Section 2.2; Supplementary Table S5), so the measurement inherits that curation and should be

read as a property of an explicitly stated absorption model rather than as an independent pharmacokinetic measurement.

### 3.5 Semi-independent audit: DDInter-derived set (92 cases)

On the 92-case audit set, `full_system` reached 0.674 exact accuracy (62/92; Wilson CI 0.573–0.761) and kappa 0.428 against the absorption-corrected gold (specificity 0.955). Only part of that set is independent of the system’s own absorption assumption: restricting the analysis to the 62 cases kept at the third-party DDInter severity gives 34 of 62 (0.548), which is the figure free of the system’s assumption and should be read as the audit-set headline rather than 0.674. Against the uncorrected DDInter severity labels, agreement was 0.370 (34/92). Stratified: among the 30 absorption-corrected cases (all corrected to low), the system agreed on 28 (93%); among the 62 kept cases, it agreed on 34 (55%) (Table 5). Baselines on the corrected gold: `pure_llm` 0.630, `lightrag` 0.591, `naive_rag` 0.535. McNemar tests on the audit set were significant for `full_system` versus `lightrag` (8 versus 1;  $p = 0.039$ ), but not versus `naive_rag` (27 versus 17;  $p = 0.174$ ) or `pure_llm` (18 versus 15;  $p = 0.728$ ). We also report, symmetrically, that on this set the two ungrounded baselines exceed the system on high-risk sensitivity (`pure_llm` and `naive_rag` 0.848 each, `lightrag` 0.264, versus 0.320 for the cascade), so the cascade’s advantage here is in specificity and exact accuracy, not in sensitivity. **That comparison is partly confounded, and we quantify the confound rather than leave it implicit.** Of the 1,307 evidence chunks, 500 are DDInter 2.0 severity statements, and the audit gold for the 62 third-party-graded cases is the DDInter value itself. Running the deposited BM25 retrieval with the deposited parameters, the top-3 passages contain the pair’s own DDInter statement for 19 of those 62 cases (`scripts/naive_rag_leakage_audit.py`); for those

cases naive\_rag retrieves the gold label. Its exact accuracy on that subset therefore cannot be read as independent evidence, and the same applies in weaker form to lightrag, whose knowledge-graph profiles are built from the same curated corpus. The deterministic cascade does not query that corpus at inference time. We report the baseline comparison because it was pre-specified, and the leakage audit so that its weight can be judged correctly.

High-risk sensitivity on the audit set was 0.320 (8 of 25 gold-high cases; CI 0.172–0.516), the lowest of the four methods (pure\_llm and naive\_rag 0.848, lightrag 0.264). This is the single most important limitation of the current system and is reported deliberately. The 30 disagreements with the audit gold were directionally consistent with the absorption-first design: 17 gold-high pairs were downgraded (8 to low, 9 to medium), 9 gold-medium pairs to low, 2 gold-medium pairs upgraded to high, and 1 gold-low pair each to medium and high (Supplementary Table S2). The four over-alerts (pairs graded high or medium by the system against a lower gold) were apraclonidine × midazolam, apraclonidine × loperamide, tacrolimus × ibuprofen, and bromfenac × enalapril.

What the system could miss. The 17 downgraded gold-high pairs include six pairs where systemic-route reasoning is weak on ophthalmic grounds (for example aflibercept × leflunomide, with no plausible topical systemic pathway). The remaining downgrades are clinically sensitive: betaxolol × verapamil and betaxolol × diltiazem (selective beta-blockade plus calcium-channel blockade), timolol × epinephrine, scopolamine or homatropine × topiramate (anticholinergic burden), and flurbiprofen × methotrexate. That last pair deserves comment because it illustrates how the absorption model interacts with the third-party labels. An earlier analysis of this registry tiered flurbiprofen high and

therefore traced the divergence to a matrix cell graded below its gold. Reviewing the assignment showed it to be a data-entry error: flurbiprofen sodium ophthalmic is formulated at 0.03%, the lowest concentration of any ophthalmic NSAID in the registry, the other four ophthalmic NSAIDs are all tiered low, and the curation file recorded low for the same two evidence sources. Correcting the tier to low (Section 4.4) moves this pair and three related anticoagulant pairs further from the DDInter-derived gold rather than closer to it, because the systemic-route label is high while a 0.03% topical formulation cannot plausibly reach systemic exposure. We report the corrected direction rather than the original one: the divergence here is between a systemic-route label and a low-absorption ophthalmic agent, not between a matrix cell and its gold. For a screening tool, this false-negative profile is the clinically critical number, and it defines the calibration priorities in Section 4.5.

### **3.6 Reasoning-model baselines under an adequate output budget**

Because the primary comparison used a single non-reasoning model with a 10-token output cap, we repeated the three LLM baselines under conditions that remove that constraint, and report both settings rather than replacing one with the other. The re-run used a reasoning model (glm-5.3-flash), a 3000-token budget, a fixed temperature of 0.0 with five repeated samples per case, and a pre-specified parsing rule (exact match first, then substring matching in the order high > medium > low, else unknown). The system prompt, retrieval settings and user-prompt templates were unchanged, so the two settings differ only in model, output budget and sampling. All 3,150 calls completed with no errors and no unparseable responses.

608 Accurate label agreement improved. On the 118-case set, pure\_llm rose from 0.475 to  
 609 0.576 and naive\_rag from 0.415 to 0.585; lightrag was unchanged at 0.737. **High-risk**  
 610 **sensitivity, however, fell sharply**, and specificity rose to approximately 1.000:

| method      | Acc3 (10-<br>token) | Acc3 (3000-<br>token) | sens (10-<br>token) | sens (3000-<br>token) | spec (3000-<br>token) |
|-------------|---------------------|-----------------------|---------------------|-----------------------|-----------------------|
| pure_llm    | 0.475               | 0.576                 | 0.771               | 0.250                 | 1.000                 |
| naive_rag   | 0.415               | 0.585                 | 0.757               | 0.214                 | 0.989                 |
| lightrag    | 0.737               | 0.737                 | 0.779               | 0.536                 | 1.000                 |
| full_system | 0.907               | 0.907                 | 0.964               | 0.964                 | 0.989                 |

611 The mechanism is a shift in the predicted label distribution rather than an improvement in  
 612 discrimination. Under the larger budget the baselines converge on a conservative policy:  
 613 pure\_llm assigns high risk to 6% of cases where the gold contains 24%, and on the audit  
 614 set to 2% where the gold contains 27%. Because both gold sets are dominated by low and  
 615 medium grades, this conservatism inflates exact accuracy while the baselines identify 6 to  
 616 15 of the 28 gold-high cases (the cascade identifies 27 of them). We therefore read the  
 617 accuracy increase as an artefact of the decision rule the model adopts when given room to  
 618 reason, not as evidence that reasoning improves ophthalmic DDI screening; the  
 619 sensitivity column is the informative one.

620 Two further results from this re-run bear on the interpretation of the primary comparison.  
 621 First, the ordering between retrieval methods reverses: under the 10-token cap naive\_rag  
 622 scored below pure\_llm (0.415 versus 0.475), and we previously attributed that to retrieval  
 623 noise, but with an adequate budget naive\_rag scores at or above pure\_llm on both sets

(0.585 versus 0.576; 0.728 versus 0.609). The earlier gap was a property of the constrained setting, and the claim that keyword retrieval injects noise in this domain is withdrawn. Second, on the audit set the adequate-budget naive\_rag baseline reaches 0.728 exact accuracy and 0.400 high-risk sensitivity, both above the cascade (0.674 and 0.320); we report this because it is the one comparison in which a baseline outperforms the system, and it is not explained by the conservatism described above. Its sensitivity is also unstable across the five samples (0.280 to 0.480), whereas the cascade returns the same grade every time.

Sample-to-sample variation was itself informative. At a fixed temperature of 0.0 the baselines still varied by 0.013 to 0.034 in exact accuracy across the five repeats, which is why the reported figures use a majority vote over five samples rather than a single call; the cascade’s SD remained 0.000.

### 3.7 Determinism and temperature sensitivity

Full\_system is deterministic: all five temperature conditions returned identical predictions (SD = 0.000 for every metric), a determinism check rather than a robustness claim. Figure 5 shows the per-layer latency trace for the worked example, in which the whole cascade completes in under 2 ms with no model call. The LLM baselines varied across conditions on the L1 set: lightrag accuracy between 0.720 and 0.771 (SD 0.018), pure\_llm between 0.449 and 0.517 (SD 0.024), and naive\_rag between 0.390 and 0.432 (SD 0.014), reflecting genuine temperature sensitivity in the LLM baselines. The same pattern held on the audit set, where the LLM baselines varied (SD 0.008–0.039) and the deterministic cascade did not. Because the five conditions differ only in temperature, these standard deviations quantify sampling instability rather than seed variance. The

deterministic path exercised here is the one distributed in the release described in Section 2.7.

### 3.8 Expert grade review (two ophthalmologists, 12 cases)

Two ophthalmology experts reviewed 12 cases under the grade-review protocol of Section 2.5, with the proposed gold and the system grade both visible (Table 6; Supplementary Table S3). Inter-rater agreement was 91.7% (11 of 12; kappa 0.840, 95% bootstrap CI 0.526–1.000; Figure 3). Expert–system agreement, computed against the grades the experts were shown, was 16.7% for expert 1 (2 of 12; kappa –0.176) and 8.3% for expert 2 (1 of 12; kappa –0.257). The disagreement was systematic and one-directional: expert 1 rated 10 of 12 cases one tier lower than the grade shown and none higher; expert 2 rated 10 lower and 1 higher. Neither expert rated any of the four system-high cases as high (0 of 4). One case was fully concordant between both experts, the grade shown and the gold (R07 levofloxacin × sotalol). The sole inter-expert disagreement was R08 (betaxolol × verapamil), graded medium by expert 1 and high by expert 2. The maximum achievable unweighted kappa given the marginal distributions is 0.294 for expert 1 and 0.314 for expert 2, so the low observed kappas partly reflect margin incompatibility; the quadratic-weighted kappas (–0.630 and –0.793) place the divergence at roughly one tier.

**Three cases carried labels that had just been corrected, and we report what the experts saw.** The questionnaire showed the experts the corrected gold, not the pre-correction DDInter value, for R01-R03. The three cases, with both values, are:

| case | DDInter<br>original | shown to<br>experts   | expert 1 | expert 2 | gold retained |
|------|---------------------|-----------------------|----------|----------|---------------|
| R01  | high                | medium<br>(corrected) | low      | low      | yes           |
| R02  | medium              | high<br>(corrected)   | medium   | medium   | yes           |
| R03  | low                 | medium<br>(corrected) | low      | low      | yes           |

For R02 and R03 both experts independently rated the *pre-correction* grade, and the gold was kept at the corrected value; that is, the corrections are not reflected in the experts' answers. The corrections were made before the review, and the procedural lock described in Section 2.3 applies to the 40-case frozen blind subset, which does not include these cases, so there is no conflict with the lock; but the sequencing is stated here rather than left for a reader to reconstruct from the deposited design file.

The experts disagreed with the gold labels to the same degree as with the system: expert 1 versus gold kappa 0.020 (agreement 33.3%) and expert 2 versus gold kappa -0.091 (agreement 25.0%). The two comparisons are not independent evidence — the questionnaire showed both the proposed gold and the system grade, and the system reproduced its gold on all 12 cases, so the expert–gold and expert–system comparisons measure the same disagreement twice. What the exercise supports is only that the experts did not endorse the assigned grades, not that the system and its gold were separately validated. Consistent with this, the experts explicitly noted that they judged from clinical

practice without mechanism-level assessment, and that ophthalmic drops are not routinely screened for systemic DDI in their practice. We interpret this as a perspective gap between clinical prior and evidence-based grading, reported transparently; with  $n = 12$  and near-zero kappa, the gap cannot be statistically distinguished from chance-level agreement, and the small sample bounds all expert-based conclusions.

### 3.9 Limitations

First, the L1 gold shares literature sources with the rules (homogeneous construction) and was author-annotated rather than expert-adjudicated; only the 40 frozen blind cases are independent in a procedural sense, and only 20 of these (blind 1) were free of development exposure. Second, 78 of 118 L1 cases overlap the development-phase validation set by case identity (gold independent, cases not entirely novel). Third, the audit gold applies the system's absorption assumption to 30 of 92 cases; against unadjusted DDInter severity, agreement drops to 0.370, and audit-set high-risk sensitivity is 0.320. Fourth, the 12-case expert exercise is a grade review, not a blinded validation of grading accuracy: both the proposed gold and the system grade were visible to the experts, they did not review mechanism evidence, the cases were selected by the study team, and expert 2's ratings were transcribed by the corresponding author from written responses rather than from a scan. Fifth, headline results reflect the final engine on the final (errata-corrected) gold; earlier engine versions scored higher on the same cases (0.992), and the drift is disclosed rather than hidden. Sixth, evidence for ophthalmic-specific DDI outcomes is sparse, so a substantial part of the rule layer is mechanism-derived. These constraints bound the strength of the accuracy claims and are reported to support a fair assessment.

## 4. Discussion

### 4.1 Principal findings

The grade review in Section 3.8 is the source for the expert comparison discussed here: the same ophthalmic–systemic pair receives three different risk assignments depending on who is judging, and this study places all three on common ground. Systemic-route database severity and absorption-aware assessment disagree for 75.3% of the 1,933 registry-matched pairs that carry a DDInter rating, and that divergence is dominated by a single mechanism: low topical systemic absorption accounts for 80.5% of downgrades. Evidence-graded rules and expert judgment also disagree, and not because of the rules — two ophthalmologists agreed closely with each other (kappa 0.840) yet rated 10 of 12 cases one tier lower than the grade they were shown, and their agreement with the assigned grades was near zero for both experts (−0.176 and −0.257); because the questionnaire displayed both the proposed gold and the system grade and the cascade reproduced that gold on all 12 cases, this review is reported as a perspective comparison rather than an independent validation. Because both grades were visible to the experts (Section 2.5), this comparison cannot separate the two assessments; what it does establish is that the grades did not persuade them. The results therefore support absorption-first knowledge engineering as a way of making ophthalmic DDI risk explicit and individually auditable, and they locate the largest source of disagreement not in any one tool but in the absence of a shared grading standard for this route.

The quantitative performance results bound that claim rather than carry it. The cascade agreed with its evidence-graded gold on 0.907 of 118 cases (kappa 0.857) and

outperformed three LLM baselines (0.415–0.737 accuracy) with significant McNemar tests, zero variance across five temperature conditions, and millisecond inference. On the semi-independent audit set it again led on exact accuracy (0.674 versus 0.535–0.591) and specificity (0.955), but its high-risk sensitivity was the lowest of the four methods (0.320, 8 of 25), and its advantage over pure\_llm and naive\_rag on that set was not statistically significant. The procedurally frozen blind subset ( $n = 40$ ) was classified with 39 of 40 correct. Because the L1 gold shares its literature sources with the rules, the 0.907 figure is best read as the internal consistency of an explicitly evidence-graded encoding; the audit-set sensitivity is the number that bounds what the system can claim as a safety screen today.

## 4.2 What we add relative to prior work

Against general-purpose DDI databases. Existing resources such as DDInter provide comprehensive pair-level severity ratings derived from systemic administration [16,N2] but do not encode the decisive ophthalmic modifier, the magnitude of systemic absorption from topical administration, nor dose-route corrections. Our system makes this ophthalmic pharmacokinetic factor an explicit, first-class modelling component [32]. A systemic-absorption tier is assigned to every ophthalmic agent (six values in the registry, two of which the L3 constraint forces to low); an absorption-scaling step forces low-absorption pairs to low risk within the matrix layer unless a mechanism-based upgrade applies; and a hard-constraint layer encodes the absorption-first pharmacology, with the final grade taken as the maximum of the rule and matrix layers. **That maximum is the honest limitation of the design.** The determinism that makes this behaviour auditable is reported in Section 3.7. The rule layer is not absorption-gated, so 202 pairs with a low-or-

below tier receive medium or high grades through rules that require no absorption flag — all 95 high grades among them from a single rule, `nsaid_antihypertensive_antagonism`. We could have gated that rule, as the four class-matrix upgrades are gated, and we measured what that would do: gating it lowers agreement with the L1 gold from 0.907 to 0.890, and gating all four NSAID rules lowers it to 0.847. We therefore report the behaviour rather than re-engineer it to fit the gold. That the gold prefers an ungated rule is itself informative: the guideline-sourced subset was labelled from sources that reason about systemic exposure, so part of the gold encodes the same systemic-route assumption the paper argues against. Whether to gate these rules is a calibration decision that should follow outcome-linked evidence, not gold agreement, and Section 4.5 lists it first. In the 1,933-pair DDInter audit (Section 3.4), 75.3% of matched pairs were graded lower by the system, including 352 pairs downgraded from high (18.2% of all pairs), with 80.5% of downgrades attributable to low systemic absorption. The methodological implication is the one worth stating plainly: a severity rating exported from a systemic-route database is not a valid ophthalmic ground truth, and any evaluation of an ophthalmic DDI tool that treats it as one measures the distance between two assumptions rather than the tool's accuracy. That is why this study reports agreement with unadjusted systemic-route labels (0.210 and 0.443 on the development and external sets, Section 3.2) as a quantification of the adjustment itself rather than as a performance figure.

Against LLM and retrieval-augmented approaches. LLM screening is non-deterministic and difficult to audit, and retrieval-augmented variants inherit the biases of their corpora [33,34]. Machine-learning predictors built on molecular structure [35] or knowledge-graph representations [36] instead target systemic-route interactions and likewise omit

route-specific absorption. In our evaluation the graph-context baseline (named lightrag; Section 2.4) captured only part of the gain even though all LLM baselines received an identical system prompt containing the key absorption facts, and under an adequate output budget the baselines converged on a conservative policy that assigns high risk to 6% of cases where the gold contains 24% (Section 3.6). What separates the cascade from the baselines is therefore not label agreement but the sensitivity-specificity profile and the reproducibility of the decision. Our system is deterministic by construction — it makes no model call, so its reproducibility is a design property rather than a robustness result — and it is auditable (every decision returns matched rule IDs, decision path, and evidence citations) and fast (about 2 ms per pair on a single CPU), while retaining the full mechanism vocabulary of the knowledge base.

Systematic coverage of the overlooked space. Clinical recall of ophthalmic–systemic DDI concentrates on a small set of well-known agents, most prominently non-selective beta-blockers. Beyond this class, interactions are rarely mentioned in routine practice and are easily overlooked [37]. Our system screens all 113 ophthalmic agents against 232 systemic drugs exhaustively. It surfaces mechanism-based signals that clinicians may not spontaneously consider, including anticholinergic accumulation with tricyclic antidepressants, additive carbonic-anhydrase inhibition, CYP3A4-mediated exposure changes, and weak-corticosteroid potency boundaries. Each signal is traceable to verified evidence.

Validation discipline. Beyond absolute performance, we contribute a four-phase evaluation protocol with explicit disclosure of its own weaknesses. The protocol includes a procedurally frozen blind subset (SHA-locked), a third-party audit component (DDInter

severity levels) with absorption-correction strata reported separately, a five-temperature protocol separating deterministic and stochastic methods, and full reporting of all four datasets including the unadjusted-gold reference sets. Homogeneous-construction, overlap, and development-exposure limitations are reported transparently. These elements are rarely reported together in clinical NLP or CDSS studies [38,39].

#### **4.3 The expert–clinical gap: a prior-versus-evidence divergence**

Two independent ophthalmologists agreed closely with each other (kappa 0.840) but rated 10 of 12 cases one tier lower than both the system and its gold. Because the questionnaire displayed both grades and the system reproduced its gold on all 12 cases, the expert–gold and expert–system comparisons are not independent and neither can be read as validating the other. The exercise establishes a systematic difference between clinical prior and the assigned grades, nothing more. Three observations support this reading. First, clinically serious events from ophthalmic–systemic DDI are rare, and expert judgments rationally gravitate toward lower risk tiers; consistent with this, neither expert rated any of the four system-high cases as high. Second, the few interactions clinicians do recall are dominated by non-selective beta-blockers, which are prominently discussed in prescribing guidance; the systematic coverage described above is therefore most valuable precisely where clinical recall is weakest. Third, reports of serious adverse events attributable to ophthalmic–systemic interactions are rare in the PubMed literature, although pharmacovigilance analyses of ophthalmic agents do identify systemic signals [29,40]. High alert-override rates in computerized order entry [24] further suggest that a screening tool whose outputs sit one tier above clinical intuition will be treated as noise unless its mechanism-based justifications are presented transparently; our traceable

decision paths are designed to do exactly that. Whether the gold, the system, or the clinical prior is the better predictor of patient-relevant outcomes cannot be resolved without outcome-linked data; the honest position is that all three are hypotheses awaiting prospective calibration.

#### 4.4 Limitations

Absorption-tier assignment. The tiers are the most load-bearing variable in the cascade and the least independently documented. They are a study-team assignment (Section 2.2) with no operational measurement behind every agent, and one defect has been corrected during revision and one remains open. **Corrected:** flurbiprofen was tiered high in the registry while the curation file recorded low for the same evidence sources; at 0.03% it is the lowest-concentration ophthalmic NSAID in the registry and the other four are all tiered low, so the high tier was a data-entry error. It is now tiered low, and the registry-scale analysis in Section 3.4 reflects the corrected value. **Disclosed alternative tiers.** Thirteen further agents show the same registry-versus-curation discrepancy that led us to correct flurbiprofen (dorzolamide, brinzolamide, the timolol fixed combinations, dexamethasone, atropine among them). We adopt the registry (data/seed/entities\_a.csv) as authoritative because it is the file the engine loads at inference time; the curation file is an intermediate artefact. To let readers judge how much the registry-scale headline depends on that choice, we re-ran the 1,933-pair audit under the curation-file tiers: the downgrade share moves from 75.27% to 75.69%, and the from-high downgrades stay at 352 (scripts/audit\_ddinter\_disagreement.py --tier-override data/curation/entity\_a\_candidates.yaml; result deposited as audits/tier\_override\_scenario.json). The registry-scale conclusion is therefore robust to which of the two files is treated as authoritative. **Open:** nine of the

113 agents in the “ophthalmic topical” registry are not topical (six intravitreal anti-VEGF agents, intraocular acetylcholine, and two surgical viscoelastics), including aflibercept, which is the example given in Section 3.5 of a pair with no plausible topical systemic pathway. Because the tier drives both the class-matrix forcing and the L3 constraint, the registry-scale divergence reported in Section 3.4 should be read as a property of this stated absorption model rather than as an independent pharmacokinetic measurement; Section 3.4 gives the tier cross-tabulation so the magnitude can be judged. Correcting a second class of assignments — the tiering of non-topical agents — remains future work, and the scenario analysis deposited alongside this manuscript quantifies how the registry-scale figures move under each candidate reassignment.

Bounded clinical impact. The scarcity of direct evidence plausibly reflects that ophthalmic absorption rarely alters systemic drug handling in a clinically detectable way. PubMed yielded few dedicated ophthalmic–systemic interaction studies or case series. The evidence base is populated largely by pharmacokinetic first principles, class-level pharmacology, and extrapolation from systemic data (1,307 curated chunks, but few ophthalmic-specific outcome studies). Ocular drug transfer from systemic administration is likewise complex and incompletely characterized [28]. If the true risk gradient clusters at the low end for most ophthalmic agents, an evidence-graded system may systematically over-grade relative to clinical practice. Conversely, the audit-set sensitivity of 0.320 shows the system can also under-grade relative to an absorption-corrected gold; both directions bound the current clinical utility. At a realistic 1% prevalence of clinically significant ophthalmic–systemic interactions, the audit operating point (sensitivity 0.320, specificity 0.955) would yield a positive predictive value of

approximately 6.7%, so the system's value lies in systematic coverage and traceability rather than in screening yield.

Validation-related limitations. The L1 gold labels were constructed from the same literature sources that informed the rules (homogeneous construction); the set is a literature-derived, author-annotated proxy rather than an expert-adjudicated gold. Only the 40 procedurally frozen blind cases are independent in the procedural sense, with 20 of them free of development exposure. In addition, 78 of 118 L1 cases overlap the development-phase internal validation set by case identity. No real human expert participated in gold construction. The two-expert grade review is the only human channel. It cannot serve as an independent test of the grading: both the proposed gold and the system grade were visible to the reviewers, and the system reproduced its gold on all 12 cases, so expert-versus-gold and expert-versus-system agreement are necessarily the same comparison and the observed one-tier gap cannot be attributed to either source. What the exercise does show is that two ophthalmologists, given no mechanism evidence, did not endorse the assigned grades, and that this divergence is stable between them; with  $n = 12$  and near-chance kappa it cannot be distinguished from chance-level agreement. One expert's ratings were transcribed by the study team from written responses. Headline results reflect the final engine and the final, errata-corrected gold; the earlier 0.992 snapshot is disclosed in the process audit. These constraints bound the strength of the accuracy claims.

#### 4.5 Future directions

Four directions follow directly, with the first taking precedence. **Registry correction:** flurbiprofen has been corrected in this revision; the remaining work is to re-assign the

888 tiers with a documented source and threshold for every agent and to reclassify or exclude  
889 the nine non-topical agents, then re-run the registry-scale analysis of Section 3.4. The tier  
890 table deposited with this work (Supplementary Table S5) and the scenario analysis in the  
891 deposit (audits/tier\_alternative\_scenarios.json) are the starting points. The same item  
892 covers the rule-layer question raised in Section 4.2: whether  
893 nsaid\_antihypertensive\_antagonism and the other NSAID rules should be gated on  
894 absorption, as the class-matrix upgrades are. We left them ungated because gating  
895 reduces agreement with the present gold, but that is a reason to revisit the gold and the  
896 calibration jointly rather than a reason to leave the design as it stands. Second, clinical  
897 calibration: a larger expert panel reviewing mechanism-level evidence, or outcome-linked  
898 retrospective data, could determine whether a systematic one-tier offset should be applied  
899 to specific absorption tiers or drug classes, converting the perspective gap into a  
900 calibrated prior. The audit-set false-negative pairs (notably flurbiprofen  $\times$  methotrexate  
901 and betaxolol  $\times$  calcium-channel blockers) are the first calibration targets. Third, false-  
902 negative surveillance: targeted pharmacovigilance mining (FAERS-style) and prospective  
903 monitoring of high-absorption agents would test the mechanism-derived rules against  
904 real-world outcomes and populate the sparse evidence base. Fourth, coverage and fail-  
905 safe behaviour: extension of the registry to additional ophthalmic and systemic agents,  
906 and a user-visible unassessed status for out-of-coverage pairs rather than a scored default,  
907 are planned. Fifth, scope extension: the class-matrix architecture transfers naturally to  
908 other specialty-systemic interaction domains (dermatologic, inhaled, and intranasal  
909 routes) with the same absorption-first design.

## 5. Conclusion

The system construction reported in Section 3.1 yields a cascade whose agreement with the literature-derived gold is quantified in Section 3.3. We developed OphDDI-CDS, an absorption-first deterministic cascade for ophthalmic–systemic DDI screening (40 mechanism rules; an absorption-scaled class matrix; a hard-constraint layer encoding topical systemic absorption), and used it to compare three ways of assigning risk to the same drug pairs. Against DDInter severity, absorption-aware assessment graded 75.3% of all 1,933 registry-matched pairs lower, with 80.5% of that shift attributable to low topical systemic absorption; a two-expert grade review likewise found systematic disagreement with the assigned grades, independently of the system only in the sense that neither the gold nor the system grade persuaded the reviewers (Section 2.5 limits what this comparison can establish). The cascade itself agreed with its evidence-graded gold on 0.907 of 118 cases (kappa 0.857), with complete output traceability, outperformed three LLM baselines on exact accuracy, and on a semi-independent audit set reached 0.674 exact accuracy — falling to 0.548 on the subset free of its own absorption assumption — but the lowest high-risk sensitivity of the four methods (0.320). That sensitivity, together with a literature-derived author-annotated gold, bounds what the system can claim today. The practical conclusion is that ophthalmic DDI risk currently lacks a shared grading standard: database severity, evidence-graded rules, and clinical prior disagree systematically, and evaluation and calibration in this domain should measure that disagreement rather than assume any one of the three is correct. Prospective, outcome-linked calibration is the next step, and the false-negative pairs identified here are its first targets.

## **Keywords**

ophthalmic drug-drug interaction; clinical decision support; deterministic rule engine; systemic absorption; explainable artificial intelligence; medication safety

## **Declarations**

### **Ethics approval and consent to participate**

This study used only publicly available secondary data: the DDInter severity tables, published drug labels, DrugBank records and the peer-reviewed literature. No human participants, patient-level data, human tissue, biological specimens or clinical interventions were involved, and no identifiable information was collected. Under the Declaration of Helsinki and the Chinese Measures for Ethical Review of Life Science and Medical Research Involving Humans, secondary analysis of published data does not constitute human-subjects research requiring ethics committee review, and the study was therefore exempt from ethics approval for the retrospective data-analysis and software-development components.

The only human involvement was a professional-opinion exercise: two ophthalmologists, both clinicians from the Department of Ophthalmology of the First Affiliated Hospital of Xinjiang Medical University, independently rated 12 hypothetical drug-pair vignettes constructed by the study team. The vignettes contained no patient data and no identifiable information; the experts provided their ratings as professional opinion and consented to their anonymised ratings being used in research and publication. One of the experts was a member of the research team of a separately approved study at the same department (ethics approval K202509-29, “Preliminary study of a mobile-device-based method for

refractive error detection”), which is the source of the institutional link between that approval and the present work; K202509-29 does not itself cover this study. The document is included in the supplementary material only to evidence that relationship, and we ask the editor to treat this study as exempt rather than as covered by that approval. The system is a research decision-support tool and is not intended to replace clinical judgment (Section 2.6).

#### **Consent for publication**

Not applicable. This study did not involve data from any individual person.

#### **Availability of data and materials**

All datasets supporting the conclusions of this article are deposited in a public repository and are also included as additional files.

**Repository.** <https://github.com/wanguoqing932/ophddi-cds> (release tag v1.1-revision3, archived at DOI {DOI}; accessible without an account). The deposit contains the entity registries (113 ophthalmic, 232 systemic agents) and the 1,307-chunk evidence base; all gold files with case-level provenance (L1 118 cases, DDInter audit 92 cases, development 200 cases, external 70 cases, plus the two development-gate sets); the 12-case expert ratings together with the review instrument and scanned questionnaires; the complete prediction files (9,600 rows, covering 4 methods x 5 temperature conditions x 4 datasets) and the evaluation and statistics scripts; the exact prompt templates; the DDInter severity tables used for the registry-scale audit; a per-agent absorption tier table with a source column for each agent; the path-attribution audit reporting which layer produced each grade; and SHA-256 fingerprints for the engine, rules and gold versions. The

977 deterministic cascade requires no API key or network access, so the pipeline can be run  
978 end to end from the deposit; re-running the LLM baselines requires credentials for the  
979 model API.

980 **Software.** Project name: OphDDI-CDS (Python package `ophthalmic_ddi_cds_agent`).  
981 Project home page: <https://github.com/wangguoqing932/ophddi-cds>. Archived version:  
982 the release corresponding to this manuscript is tagged v1.1-revision3 and is archived at  
983 DOI {DOI} (Zenodo). The archive contains the full source, configurations, gold sets,  
984 9,600 prediction rows, the 3,150 reasoning-model responses, the statistics module and  
985 every audit output, so the reported figures can be regenerated without network access.  
986 Operating system(s): platform independent. Programming language: Python 3.11. Other  
987 requirements: Python 3.10 or later, with dependencies pinned in `pyproject.toml`. Licence:  
988 Apache-2.0 (code) and CC BY 4.0 (original data). The bundled DDInter severity tables  
989 are third-party data and remain under DDInter's CC BY-NC-SA 4.0 terms; they are  
990 included for research reproducibility only, and reuse is subject to those terms (see the  
991 component-licensing section of the repository LICENSE).

992 The repository contains the production screening engine, the curated knowledge base, the  
993 rule-citation verification tooling and the evaluation scripts, together with a synthetic  
994 demonstration dataset. The deterministic cascade described in this paper makes no LLM  
995 calls and requires no credential or network access, so the pipeline can be executed end to  
996 end with a single command (`python scripts/demo_pipeline.py`), which reproduces the  
997 expected risk level for 25 bundled mock pairs covering all three decision paths and the  
998 unknown-drug fail-safe behaviour. A test suite (`pytest tests/`, 34 passing, 1 skipped)  
999 verifies the registry, rules and citation-verification layers.

1000 **Competing interests**

1001 The authors declare that the research was conducted in the absence of any commercial or  
1002 financial relationships that could be construed as a potential conflict of interest.

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1006 **Authors' contributions**

1007 Wang G: Conceptualization, Methodology, Software, Formal analysis, Data curation,  
1008 Visualization, Writing - original draft. Hu L: Clinical input, Data interpretation, Writing -  
1009 review & editing. Wang S: Clinical input, Data interpretation, Writing - review & editing.  
1010 Yi X: Conceptualization, Supervision, Project administration, Funding acquisition,  
1011 Resources, Writing - review & editing. All authors contributed to the article and  
1012 approved the submitted version.

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1015 **Generative AI**

1016 Generative AI tools were used during this research and manuscript preparation, with full  
1017 author review and responsibility. DeepSeek-chat (deepseek.ai) was used for LLM-  
1018 baseline evaluations, for literature-assisted drafting, and as an assistive tool in  
1019 knowledge-engineering tasks, including registry curation, evidence grading, and gold-  
1020 label annotation; Zhipu GLM-4.6V was used for visual verification of scanned expert

questionnaires and figures. Figure 1 (system architecture schematic) was created with an AI image-generation tool from a prompt defined by the authors (prompt archived in the supplementary material, figure\_prompts.md); all quantitative figures (Figures 2–5) were generated directly from system outputs with Matplotlib. All AI-assisted content was reviewed and validated by the authors, who take full responsibility for the content of this manuscript. The authors disclose that no human expert participated in gold-label construction; the 12-case expert assessment is the sole human validation channel. For clarity about the “T1 expert” referred to in the deposited design files: that label denotes the clinical opinion recorded in an earlier, separately conducted exercise by a member of the research team of the approved study cited in the Ethics statement, used only to phrase the questionnaire vignettes and to cross-check mechanism plausibility for R02 and R12. No human expert assigned any gold label in this study; every gold label is the study team’s own annotation from published sources, and the T1 comparison is reported in the deposit as provenance for how the vignettes were written, not as a source of gold.

**Table 1. Registry summary of the knowledge base (113 ophthalmic agents, 232 systemic agents).**

| entity_type      | count | with_flags | with_drugbank_ids |
|------------------|-------|------------|-------------------|
| ophthalmic_drugs | 113   | 113        | 103               |
| systemic_drugs   | 232   | 232        | 218               |

**Table 2. Rule inventory: 40 mechanism rules plus 3 patient-factor policies (43 policy entries) with severity, risk type, flags, and match semantics.**

| rule_id                   | severity | risk_type                      | entity_a_flags | entity_b_flags           | match                          |     |
|---------------------------|----------|--------------------------------|----------------|--------------------------|--------------------------------|-----|
| additive_beta_blockade    | high     | pharmacodynamic_cardiovascular | beta_blocker   | systemic_absorption_high | beta_blocker_systemic          | all |
| beta_blocker_ccb_additive | high     | pharmacodynamic_cardiovascular | beta_blocker   | systemic_absorption_high | l_type_calcium_channel_blocker | all |

|                                         |        |                                     |                            |                          |                       |     |
|-----------------------------------------|--------|-------------------------------------|----------------------------|--------------------------|-----------------------|-----|
| beta_blocker_digoxin_additive           | high   | pharmacodynamic_cardiovascular      | beta_blocker               | systemic_absorption_high | na_k_atpase_inhibitor | all |
| additive_alpha2_cns_depression          | high   | pharmacodynamic_cns                 | alpha2_agonist             | cns_depressant           | any                   |     |
| additive_anticholinergic_effects        | medium | pharmacodynamic_anticholinergic     | anticholinergic            | anticholinergic_systemic | any                   |     |
| additive_corticosteroid_iop             | medium | pharmacodynamic_ocular_hypertension | corticosteroid             | corticosteroid_systemic  | any                   |     |
| additive_nsaib_bleeding                 | medium | pharmacodynamic_hemorrhagic         | nsaib                      | anticoagulant            | any                   |     |
| sympathomimetic_alpha_interaction       | medium | pharmacodynamic_cardiovascular      | sympathomimetic            | antihypertensive         | any                   |     |
| qt_prolongation_risk                    | low    | pharmacodynamic_cardiotoxic         | fluoroquinolone            | qt_prolonging            | any                   |     |
| pga_antiplatelet_bleeding               | low    | pharmacodynamic_hemorrhagic         | prostaglandin_analog       | antiplatelet             | any                   |     |
| cai_diuretic_metabolic_acidosis         | medium | pharmacokinetic_metabolic           | cai_topical                | diuretic                 | any                   |     |
| cai_topiramate_acidosis                 | medium | pharmacokinetic_metabolic           | cai_topical                | cai_systemic             | any                   |     |
| dexamethasone_cyp3a4_induction          | medium | pharmacokinetic_cyp3a4              | corticosteroid             | systemic_absorption_high | cyp3a4_substrate      | all |
| topical_corticosteroid_cyp3a4_induction | low    | pharmacokinetic_cyp3a4              | corticosteroid             | cyp3a4_substrate         | any                   |     |
| systemic_absorption_high_risk           | low    | systemic_absorption                 | systemic_absorption_high   | beta_blocker_systemic    | any                   |     |
| systemic_absorption_moderate_risk       | low    | systemic_absorption                 | systemic_absorption_medium | narrow_therapeutic_index | any                   |     |
| p_gp_mediated_absorption                | low    | pharmacokinetic_transporter         | p_gp_substrate             | p_gp_inhibitor           | any                   |     |
| calcineurin_inhibitor_cyp3a4            | low    | pharmacokinetic_cyp3a4              | calcineurin_inhibitor      | cyp3a4_inhibitor         | any                   |     |
| macrolide_cyp3a4_inhibition             | low    | pharmacokinetic_cyp3a4              | macrolide                  | cyp3a4_substrate         | any                   |     |
| azole_cyp3a4_inhibition                 | low    | pharmacokinetic_cyp3a4              | azole                      | cyp3a4_substrate         | any                   |     |
| sympathomimetic_maoi_hypertension       | high   | pharmacodynamic_cardiovascular      | sympathomimetic            | maoi                     | any                   |     |
| beta_blocker_beta2_agonist_antagonism   | high   | pharmacodynamic_respiratory         | beta_blocker               | beta2_adrenergic_agonist | any                   |     |
| immunosuppressant_nsaib_nephrotoxicity  | medium | pharmacodynamic_renal               | calcineurin_inhibitor      | cox1_cox2_inhibitor      | any                   |     |
| fluoroquinolone_corticosteroid_tendon   | low    | pharmacodynamic_musculoskeletal     | fluoroquinolone            | corticosteroid_systemic  | any                   |     |
| nsaib_antihypertensive_antagonism       | high   | pharmacodynamic_cardiovascular      | nsaib                      | antihypertensive         | any                   |     |
| nsaib_diuretic_antagonism               | medium | pharmacodynamic_cardiovascular      | nsaib                      | ncc_inhibitor            | nkcc2_inhibitor       | any |

|                                  |        |                                 |                            |                          |                                       |     |
|----------------------------------|--------|---------------------------------|----------------------------|--------------------------|---------------------------------------|-----|
| m                                | um     | ovascular                       |                            |                          |                                       | y   |
| antibiotic_warfarin_enhancement  | low    | pharmacokinetic_anticoagulation | antibiotic                 | narrow_therapeutic_index | vitamin_k_epoxide_reductase_inhibitor | all |
| beta_blocker_cyp2d6_metabolism   | medium | pharmacokinetic_cyp2d6          | beta_blocker               | cyp2d6_inhibitor         | any                                   |     |
| nsaid_acei_arb_antagonism        | medium | pharmacodynamic_cardiovascular  | nsaid                      | ace_inhibitor            | angiotensin_ii_receptor_antagonist    | any |
| corticosteroid_nti_interaction   | medium | pharmacodynamic_endocrine       | corticosteroid             | narrow_therapeutic_index | any                                   |     |
| sympathomimetic_tca_hypertension | high   | pharmacodynamic_cardiovascular  | sympathomimetic            | net_inhibitor            | any                                   |     |
| anticholinergic_qt_prolongation  | medium | pharmacodynamic_cardiovascular  | anticholinergic            | qt_prolonging            | any                                   |     |
| moderate_abs_antiplatelet        | low    | pharmacodynamic_hemorrhagic     | systemic_absorption_medium | antiplatelet             | any                                   |     |
| moderate_abs_beta_blocker        | medium | pharmacodynamic_cardiovascular  | systemic_absorption_medium | beta_blocker_systemic    | any                                   |     |
| moderate_abs_anticoagulant       | low    | pharmacodynamic_hemorrhagic     | systemic_absorption_medium | anticoagulant            | any                                   |     |
| moderate_abs_cyp2d6              | low    | pharmacokinetic_cyp2d6          | systemic_absorption_medium | cyp2d6_substrate         | any                                   |     |
| moderate_abs_diuretic            | low    | pharmacokinetic_metabolic       | systemic_absorption_medium | diuretic                 | any                                   |     |
| alpha2_agonist_antiplatelet      | low    | pharmacodynamic_hemorrhagic     | alpha2_agonist             | antiplatelet             | any                                   |     |
| low_abs_antiplatelet             | low    | pharmacodynamic_hemorrhagic     | systemic_absorption_low    | antiplatelet             | any                                   |     |
| low_abs_anticoagulant            | low    | pharmacodynamic_hemorrhagic     | systemic_absorption_low    | anticoagulant            | any                                   |     |

**Table 3. Four-method performance on the two validation sets (mean over five temperature conditions).**

| dataset_id | setting    | method      | n   | sensitivity_high | specificity_high | f1_high | accuracy_3level | cohens_kappa |
|------------|------------|-------------|-----|------------------|------------------|---------|-----------------|--------------|
| blind_11   | 10-token   | pure_llm    | 118 | 0.771            | 0.542            | 0.476   | 0.475           | 0.256        |
| blind_11   | 10-token   | naive_rag   | 118 | 0.757            | 0.462            | 0.434   | 0.415           | 0.182        |
| blind_11   | 10-token   | lightrag    | 118 | 0.779            | 0.936            | 0.785   | 0.737           | 0.603        |
| blind_11   | 10-token   | full_system | 118 | 0.964            | 0.989            | 0.964   | 0.907           | 0.857        |
| blind_11   | 3000-token | pure_llm    | 118 | 0.250            | 1.000            | 0.400   | 0.576           | 0.333        |
| blind_11   | 3000-token | naive_rag   | 118 | 0.214            | 0.989            | 0.343   | 0.585           | 0.341        |

|                  |            |             |     |       |       |       |       |       |
|------------------|------------|-------------|-----|-------|-------|-------|-------|-------|
| blind_l1         | 3000-token | lightrag    | 118 | 0.536 | 1.000 | 0.698 | 0.737 | 0.587 |
| blind_v3_ddinter | 10-token   | pure_llm    | 92  | 0.848 | 0.704 | 0.642 | 0.630 | 0.396 |
| blind_v3_ddinter | 10-token   | naive_rag   | 92  | 0.848 | 0.579 | 0.570 | 0.535 | 0.290 |
| blind_v3_ddinter | 10-token   | lightrag    | 92  | 0.264 | 0.949 | 0.377 | 0.591 | 0.220 |
| blind_v3_ddinter | 10-token   | full_system | 92  | 0.320 | 0.955 | 0.444 | 0.674 | 0.428 |
| blind_v3_ddinter | 3000-token | pure_llm    | 92  | 0.080 | 1.000 | 0.148 | 0.609 | 0.317 |
| blind_v3_ddinter | 3000-token | naive_rag   | 92  | 0.400 | 1.000 | 0.571 | 0.728 | 0.553 |
| blind_v3_ddinter | 3000-token | lightrag    | 92  | 0.240 | 1.000 | 0.387 | 0.674 | 0.428 |

Note: exact accuracy is on the three-level scale; 10-token denotes the primary comparison (non-reasoning model, 10-token cap, one call per case); 3000-token denotes the re-run (reasoning model glm-5.3-flash, 3000-token budget, temperature 0.0, majority vote over five repeated samples). full\_system is deterministic and therefore identical under both settings (SD = 0). Sensitivities are computed against each dataset's gold; the 3000-token baselines assign high risk far more rarely, which raises specificity to ~1.000 while depressing sensitivity.

**Table 4. Gold-level distribution of the validation sets.**

| dataset_id       | gold_level | count | percentage |
|------------------|------------|-------|------------|
| blind_l1         | high       | 28    | 23.7%      |
| blind_l1         | medium     | 51    | 43.2%      |
| blind_l1         | low        | 39    | 33.1%      |
| blind_v3_ddinter | high       | 25    | 27.2%      |
| blind_v3_ddinter | medium     | 19    | 20.7%      |
| blind_v3_ddinter | low        | 48    | 52.2%      |
| blind_l1         | total      | 118   | 100.0%     |
| blind_v3_ddinter | total      | 92    | 100.0%     |
| all              | total      | 210   | -          |

**Table 5. DDInter audit strata and disagreement composition (92 cases).**

| stratum                                      | n  | system agreement |
|----------------------------------------------|----|------------------|
| All 92 (vs absorption-corrected gold)        | 92 | 62 (0.674)       |
| All 92 (vs uncorrected DDInter severity)     | 92 | 34 (0.370)       |
| Absorption-corrected cases (gold set to low) | 30 | 28 (0.933)       |

|                                                   |    |            |
|---------------------------------------------------|----|------------|
| Kept cases (third-party severity, gold = DDInter) | 62 | 34 (0.548) |
|---------------------------------------------------|----|------------|

| disagreement direction (vs corrected gold) | n  |
|--------------------------------------------|----|
| high->low                                  | 8  |
| high->medium                               | 9  |
| medium->low                                | 9  |
| medium->high                               | 2  |
| low->medium                                | 1  |
| low->high                                  | 1  |
| total disagreements                        | 30 |

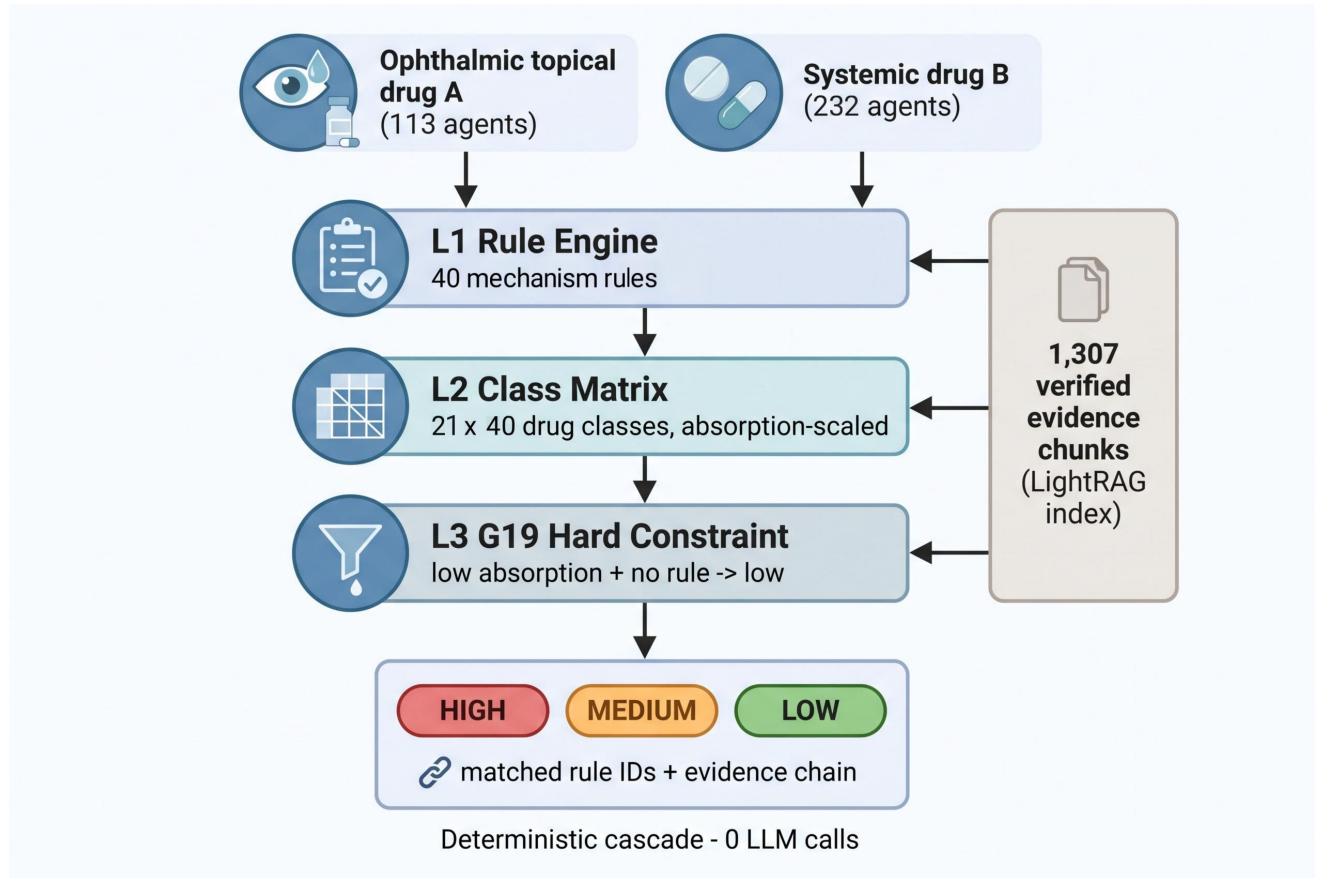
Note: values reflect the corrected flurbiprofen absorption tier. The four over-alerts (system grade above gold) are apraclonidine x midazolam, apraclonidine x loperamide, tacrolimus x ibuprofen and bromfenac x enalapril. The 62-case "kept" stratum is the subset whose gold is the third-party DDInter severity, so it is the only stratum free of the system's own absorption assumption; the 0.548 figure there should be read as the audit-set headline rather than 0.674. Of those 62 cases, the deposited BM25 retrieval returns the pair's own DDInter statement within the top-3 passages for 19, so the baseline figures on this stratum are partly confounded (see scripts/naive\_rag\_leakage\_audit.py).

**Table 6. Expert assessment: per-case ratings (gold, system, expert 1, expert 2).**

| case | ophthalmic      | systemic      | gold shown | system shown | expert1 | expert2 |
|------|-----------------|---------------|------------|--------------|---------|---------|
| R01  | Ketorolac       | Enoxaparin    | medium     | medium       | low     | low     |
| R02  | Brimonidine     | Tramadol      | high       | high         | medium  | medium  |
| R03  | Dexamethasone   | Rifampicin    | medium     | medium       | low     | low     |
| R04  | Atropine        | Tramadol      | high       | high         | medium  | medium  |
| R05  | Scopolamine     | Amitriptyline | high       | high         | medium  | medium  |
| R06  | Atropine        | Oxycodone     | high       | high         | medium  | medium  |
| R07  | Levofloxacin    | Sotalol       | low        | low          | low     | low     |
| R08  | Betaxolol       | Verapamil     | medium     | medium       | medium  | high    |
| R09  | Ketorolac       | Rivaroxaban   | medium     | medium       | low     | low     |
| R10  | Fluorometholone | Prednisone    | low        | medium       | low     | low     |
| R11  | Fluorometholone | Glimepiride   | low        | medium       | low     | low     |
| R12  | Ketorolac       | Warfarin      | medium     | medium       | low     | low     |

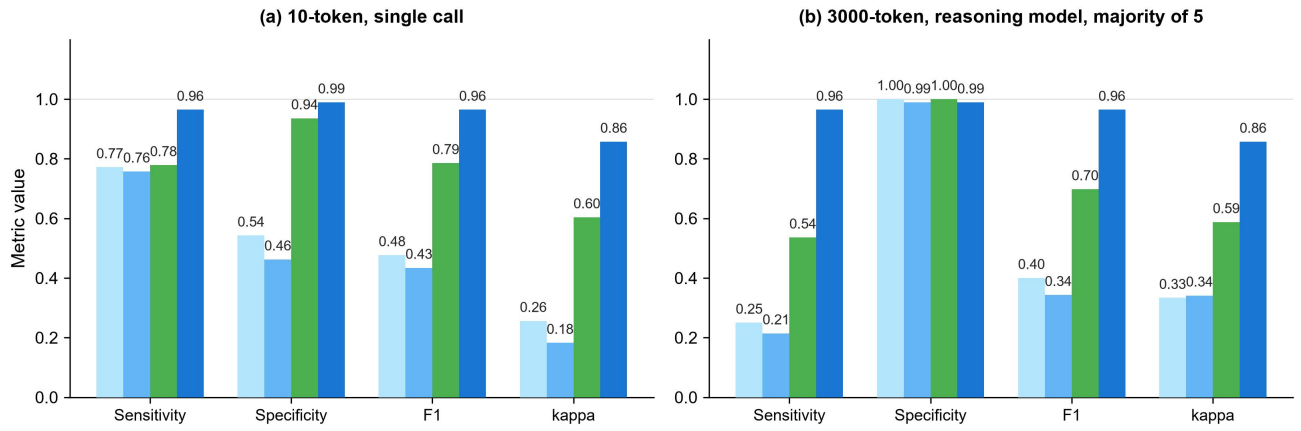
Note: system shown is the system grade displayed on the questionnaire the experts graded, which is the correct comparator for expert-system agreement. It is the value the engine returned at the time for R01-R09 and R12; for R10 and R11 the engine has since been changed (the displayed grade was medium, the current engine returns low). Expert

values are transcribed from the returned questionnaires: expert 1's form is deposited as a scan, expert 2's responses were returned in text. R03 is recorded here as low/low; an earlier version of this table listed medium/medium in error, and the kappa values reported in the manuscript are computed from the values in this table.

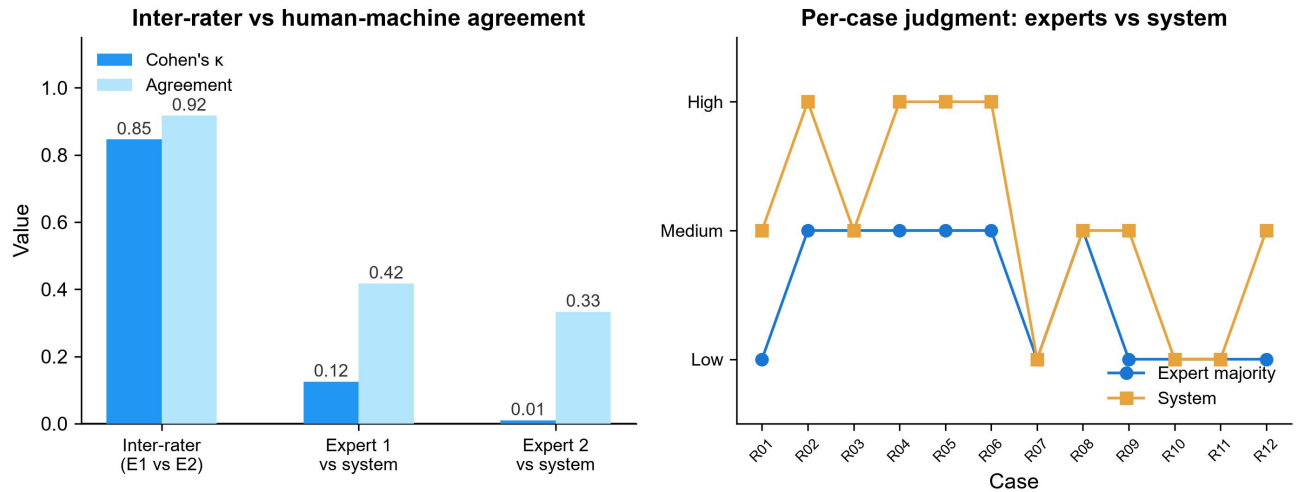


**Figure 1. System architecture of the ophthalmic drug–drug interaction clinical decision support system.**

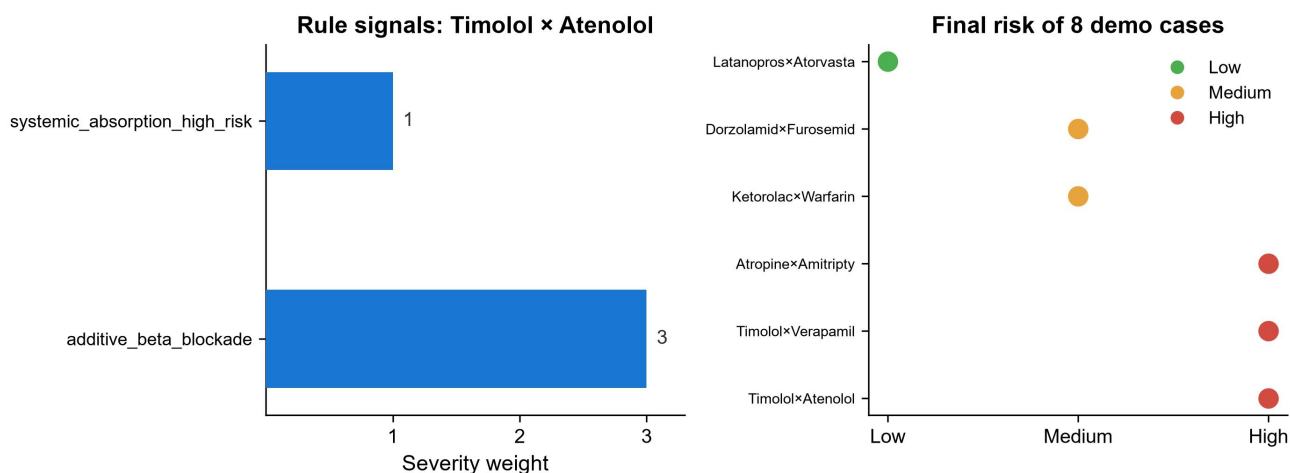
Pure LLM Naive RAG LightRAG Full system  
L1 literature-consensus benchmark (118 cases): baselines under two output budgets



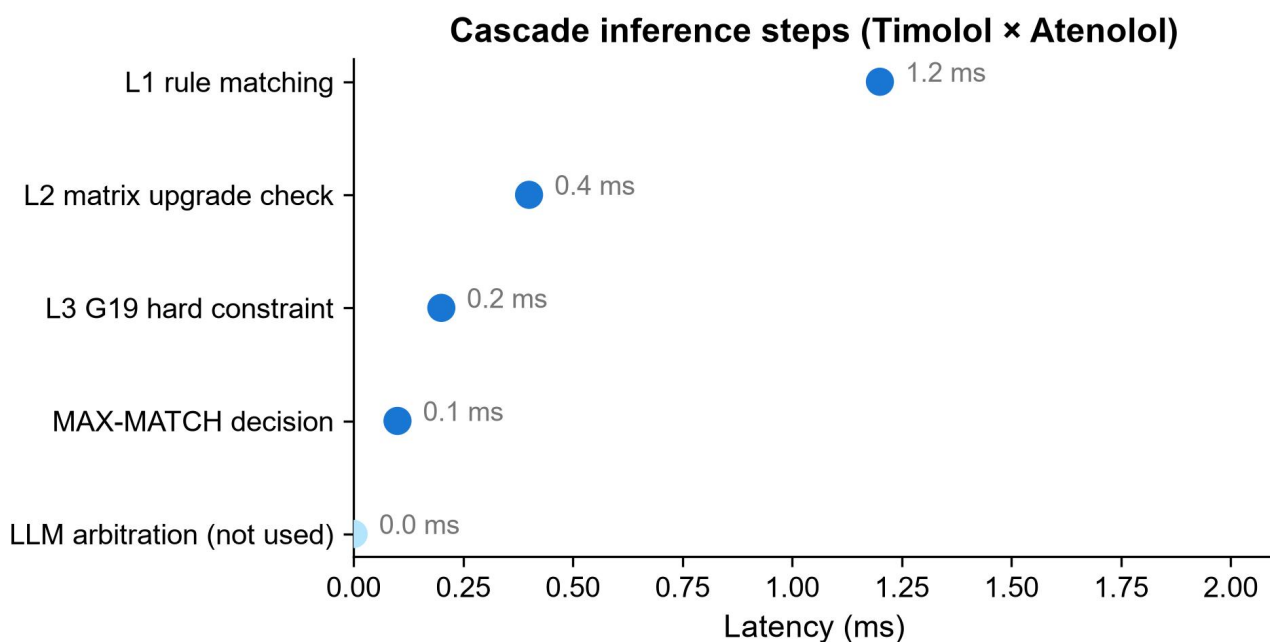
**Figure 2. Performance comparison across four methods and two validation sets.**



**Figure 3. Expert-system agreement and inter-rater reliability (two clinical experts, 12 cases).**



**Figure 4. Worked demo case: ophthalmic x systemic risk evaluation (Timolol x Atenolol).**



**Figure 5. Step-by-step reasoning trace across the cascade layers (Timolol x Atenolol).**

## References

- [1] Alvan G, Calissendorff B, Seideman P, Widmark K, Widmark G. Absorption of ocular timolol. Clin Pharmacokinet. 1980;5(1):95-100. doi: 10.2165/00003088-198005010-00004

1078 [2] Rait JL. Systemic effects of topical ophthalmic beta-adrenoceptor antagonists. Aust N  
1079 Z J Ophthalmol. 1999;27(1):57-64. doi: 10.1046/j.1440-1606.1999.00145.x

1080 [3] Yoshioka K, Kawasaki T. Atrioventricular block caused by carteolol eye drops  
1081 administered to unilateral eye or bilateral eyes: a case report. Cureus. 2025. doi:  
1082 10.7759/cureus.90018

1083 [4] Bollinger A, Jeiziner C, Meyer zu Schwabedissen H, Hersberger K, Allemann S,  
1084 Stäuble C. Severe systemic adverse reactions to ophthalmic timolol in a CYP2D6  
1085 homozygous \*4 allele carrier: a case report. Pharmacogenomics. 2023. doi: 10.2217/pgs-  
1086 2023-0122

1087 [5] Pringle SD, MacEwen CJ. Severe bradycardia due to interaction of timolol eye drops  
1088 and verapamil. BMJ. 1987;294(6565):155. doi: 10.1136/bmj.294.6565.155-a

1089 [6] Rains J, Kesterson J. Ocular timolol as the causative agent for symptomatic  
1090 bradycardia in an 89-year-old female. Am J Emerg Med. 2021. doi:  
1091 10.1016/j.ajem.2020.09.030

1092 [7] Abbas S, Hamadani S, Ahmad U, Desai A, Kitchloo K. Ophthalmic timolol and  
1093 hospitalization for symptomatic bradycardia and syncope: a case series. Cureus. 2020. doi:  
1094 10.7759/cureus.7270

1095 [8] Diggory P, Franks W. Glaucoma therapy may take your breath away. Age Ageing.  
1096 1997;26(2):63-67. doi: 10.1093/ageing/26.2.63

1097 [9] Kiryazov K, Stefova M, Iotova V. Can ophthalmic drops cause central nervous  
1098 system depression and cardiogenic shock in infants? *Pediatr Emerg Care*. 2013. doi:  
1099 10.1097/pec.0b013e3182aa1384

1100 [10] Rangan C, Everson G, Cantrell F. Central alpha-2 adrenergic eye drops. *Pediatr*  
1101 *Emerg Care*. 2008. doi: 10.1097/pec.0b013e3181668aee

1102 [11] Hoffmann U, Kuno S, Franke G, Fusch C, Haas J. Adrenoceptor agonist poisoning  
1103 after accidental oral ingestion of brimonidine eye drops. *Pediatr Crit Care Med*. 2004. doi:  
1104 10.1097/01.pcc.0000123548.73664.18

1105 [12] Gunes A, Balik H, Yel S, Kocamaz H, Bosnak M. Respiratory depression after  
1106 accidental nasal ingestion of brimonidine eye drops in an infant. *Turk J Emerg Med*. 2016.  
1107 doi: 10.1016/j.tjem.2016.02.002

1108 [13] Sacchi M, Lizzio R, Villani E, Monsellato G, Lucentini S, Cremonesi E, et  
1109 al. Medical management of pediatric glaucoma: lessons learned from randomized clinical  
1110 trials. *Graefes Arch Clin Exp Ophthalmol*. 2020. doi: 10.1007/s00417-020-04767-9

1111 [14] Xu L, Wang X, Wu M. Topical medication instillation techniques for glaucoma.  
1112 *Cochrane Database Syst Rev*. 2017;2:CD010520. doi: 10.1002/14651858.cd010520.pub2

1113 [15] Masnoon N, Shakib S, Kalisch-Ellett L, Caughey GE. What is polypharmacy? A  
1114 systematic review of definitions. *BMC Geriatr*. 2017;17(1):230. doi: 10.1186/s12877-  
1115 017-0621-2

1116 [16] Xiong G, Yang Z, Yi J, Wang N, Wang L, Zhu H, et al. DDInter: an online drug-  
 1117 drug interaction database towards improving clinical decision-making and patient safety.  
 1118 Nucleic Acids Res. 2022. doi: 10.1093/nar/gkab880

1119 [17] Percha B, Altman RB. Informatics confronts drug-drug interactions. Trends  
 1120 Pharmacol Sci. 2013;34(3):178-184. doi: 10.1016/j.tips.2013.01.006

1121 [18] Yao H, Peng Z, Huang Y, Xiao S, Wu R. Gaps in the detection of drug-drug  
 1122 interactions between antipsychotic and cardiometabolic medications: a multisource  
 1123 analysis. BMC Med. 2025. doi: 10.1186/s12916-025-04420-4

1124 [19] Singh P, Devadiga A, Pal R, Betrabet P, Sankar J, Sethi T, et al. ContraIndicator: a  
 1125 natural language processing-based approach to potential drug-drug interaction detection  
 1126 in pediatric intensive care. Clin Drug Investig. 2026. doi: 10.1007/s40261-026-01529-z

1127 [20] Robertson S, Zaragoza H. The probabilistic relevance framework: BM25 and  
 1128 beyond. Found Trends Inf Retr. 2009;3(4):333-389. doi: 10.1561/15000000019

1129 [21] Guo Z, Xia L, Yu Y, Ao T, Huang C. LightRAG: simple and fast retrieval-  
 1130 augmented generation. arXiv. 2024:arXiv:2410.05779. doi: 10.48550/arXiv.2410.05779

1131 [22] Kawamoto K, Houlihan CA, Balas EA, Lobach DF. Improving clinical practice  
 1132 using clinical decision support systems: a systematic review of trials to identify features  
 1133 critical to success. BMJ. 2005;330(7494):765. doi: 10.1136/bmj.38398.500764.8F

1134 [23] Bright TJ, Wong A, Dhurjati R, Bristow E, Bastian L, Coeytaux RR, et al. Effect of  
 1135 clinical decision-support systems: a systematic review. Ann Intern Med. 2012;157(1):29-  
 1136 43. doi: 10.7326/0003-4819-157-1-201207030-00450

1137 [24] van der Sijs H, Aarts J, Vulto A, Berg M. Overriding of drug safety alerts in  
 1138 computerized physician order entry. *J Am Med Inform Assoc.* 2006;13(2):138-147. doi:  
 1139 10.1197/jamia.M1809

1140 [25] Ray CE, Wilson GM, Hughes AM, Cunningham Goedken C, Liu EP, Fitzpatrick  
 1141 MA, et al. Alert fatigue measurement in clinical decision support: a systematic review. *J*  
 1142 *Am Med Inform Assoc.* 2026;33(8):1523-1531. doi: 10.1093/jamia/ocag064

1143 [26] Joint Formulary Committee. Glaucoma and ocular hypertension. In: British National  
 1144 Formulary. London: BMJ Group and Pharmaceutical Press; accessed July 2026. doi:  
 1145 10.18578/bnf.270842008

1146 [27] Müller L, Jensen BP, Bachmann LM, Wong D, Wells AP. New technique to reduce  
 1147 systemic side effects of timolol eye drops: the tissue press method - crossover clinical  
 1148 trial. *Clin Exp Ophthalmol.* 2020;48(1):24-30. doi: 10.1111/ceo.13642

1149 [28] Jumbe N, Miller M. Ocular drug transfer following systemic drug administration. In:  
 1150 *Ophthalmic Drug Delivery Systems.* 3rd ed. Boca Raton: CRC Press; 2021. doi:  
 1151 10.4324/9781003113423-6

1152 [29] Xu Z, Liang J, Cai W, Luo Y, Liang F, Lin R, et al. A systemic pharmacovigilance  
 1153 assessment of ophthalmic atropine: signal detection and clinical prioritization from the  
 1154 FAERS database. *Front Med (Lausanne).* 2026. doi: 10.3389/fmed.2026.1798521

1155 [30] Cohen J. A coefficient of agreement for nominal scales. *Educ Psychol Meas.*  
 1156 1960;20(1):37-46. doi: 10.1177/001316446002000104

1157 [31] Landis JR, Koch GG. The measurement of observer agreement for categorical data.  
 1158 Biometrics. 1977;33(1):159-174. doi: 10.2307/2529310

1159 [32] Chou E, Boyce RD, Balkan B, Subbian V, Romero A, Hansten PD, et al. Designing  
 1160 and evaluating contextualized drug-drug interaction algorithms. JAMIA Open.  
 1161 2021;4(1):ooab023. doi: 10.1093/jamiaopen/ooab023

1162 [33] Clusmann J, Freyer O, Ostermann M, Ferber D, Ghaffari Laleh N, Hilgers L, et  
 1163 al. Safety and security of large language models in healthcare. Nature.  
 1164 2026;656(8128):577-589. doi: 10.1038/s41586-026-10687-1

1165 [34] Leibovitch L, Ahituv A, Gorenshstein A, Aran D, Sorka M, Miron K, et  
 1166 al. Prospective evaluation of a large language model clinical decision support system in  
 1167 the emergency department. Nat Med. 2026. doi: 10.1038/s41591-026-04601-5

1168 [35] Ryu JY, Kim HU, Lee SY. Deep learning improves prediction of drug-drug and  
 1169 drug-food interactions. Proc Natl Acad Sci U S A. 2018;115(18):E4304-E4311. doi:  
 1170 10.1073/pnas.1803294115

1171 [36] Lin X, Zhang S, Li B, Liang P, Pang J. HSAF-DDI: Heterogeneous Semantic-Aware  
 1172 Drug-Drug Interaction Prediction with Hierarchical Adaptive Fusion. J Chem Inf Model.  
 1173 2026;66(16):10379-10395. doi: 10.1021/acs.jcim.6c01086

1174 [37] Edrees H, Amato MG, Wong A, Seger DL, Bates DW. High-priority drug-drug  
 1175 interaction clinical decision support overrides in a newly implemented commercial  
 1176 computerized provider order-entry system: Override appropriateness and adverse drug  
 1177 events. J Am Med Inform Assoc. 2020;27(6):893-900. doi: 10.1093/jamia/ocaa034

1178 [38] Collins GS, Moons KGM, Dhiman P, Riley RD, Beam AL, Van Calster B, et  
 1179 al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that  
 1180 use regression or machine learning methods. *BMJ*. 2024;385:e078378. doi: 10.1136/bmj-  
 1181 2023-078378

1182 [39] Vasey B, Nagendran M, Campbell B, Clifton DA, Collins GS, Denaxas S, et  
 1183 al. Reporting guideline for the early-stage clinical evaluation of decision support systems  
 1184 driven by artificial intelligence: DECIDE-AI. *Nat Med*. 2022;28(5):924-933. doi:  
 1185 10.1038/s41591-022-01772-9

1186 [40] Prasad M, Ramsey D. Adverse events reported with standard-dose and high-dose  
 1187 aflibercept: a FAERS pharmacovigilance study. *Vision*. 2026. doi:  
 1188 10.3390/vision10020018

1189 [N1] Korte JM, Kaila T, Saari KM. Systemic bioavailability and cardiopulmonary effects  
 1190 of 0.25%, 0.5%, and 1% timolol eyedrops. *Graefes Arch Clin Exp Ophthalmol*.  
 1191 2002;240(6):430-435. doi: 10.1007/s00417-002-0462-2 [N2] Tian S, Wang N, Yang Z,  
 1192 Li Y, Xiong G, Wang L, et al. DDInter 2.0: an enhanced drug-drug interaction database  
 1193 for precision medicine. *Nucleic Acids Res*. 2025. doi: 10.1093/nar/gkae726